Fraudulent Financial Reporting and the Consequences for Employees

by

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Abstract

We combine U.S. Census data with SEC enforcement actions to examine employees' outcomes, such as wages and turnover, before, during, and after periods of fraudulent financial reporting. We find that fraud firms' employees lose about 50% of cumulative annual wages, compared to a matched sample, and the separation rate is much higher after fraud periods. Yet, employment growth at fraud firms is positive during fraud periods; these firms overbuild and hire new, lower-paid employees as part of the fraud, unlike firms in distress which tend to contract. When the fraud is revealed, firms shed workers, unwinding this abnormal growth and resulting in most of the negative wage consequences. Low wage employees, though unlikely to have perpetrated the fraud, experience more severe wage losses. Other sample splits show that negative wage effects are larger in thin labor markets and for fraud firms that go bankrupt.

Keyword*:* Wages, Employment Growth, Accounting Fraud, Information Asymmetry

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1. Introduction

Fraudulent financial reporting is an important issue in the economy. Large accounting scandals occur regularly (e.g., Enron, WorldCom, Waste Management, Toshiba, Luckin Coffee, and so on), and the consequences are usually significant. For example, Karpoff et al. (2008b) find that firms lose about 29% of equity value when fraud is revealed. An extensive academic literature has also documented severe consequences of fraudulent reporting for other stakeholders, including customers, executives, and peer firms (e.g., Sadka, 2006; Desai et al., 2006; Beatty et al., 2013). However, prior papers rarely study labor market consequences, which can be large; for example, 17,000 workers lost jobs from WorldCom alone in June 2002 (Noguchi, 2002). We seek to answer two, related research questions. First, what are (the magnitudes of) the effects on labor markets and who is affected? Second, what kind of real actions do executives take along with accounting fraud that have these consequences for employees? A reasonable assumption is that executives use fraud to cover up distress, so as a corollary question, how do real actions differ in fraud cases compared with employment consequences of firm distress (e.g., restructurings or layoffs)?

Accounting fraud has two distinct features that make it likely to have consequences for employees and differ from firm distress. First, executives take real actions to complement or have undetected accounting fraud. They overinvest in physical capital during fraudulent reporting periods (McNichols and Stubben, 2008). Similarly, executives could show growth with excessive hiring while avoiding a high wage bill, e.g., by replacing high paid workers with lower paid workers. Second, employees rely on information from and about the firm when making career choices (e.g., Choi et al., 2020; deHaan et al., 2020) and are important stakeholders of the firm. If employees are misled about the prospects of the firm during the fraud, they could suffer along with other stakeholders. For example, excessive hiring could result in unaware employees suffering

later when these excesses are unwound, losing job specific capital or job hunting in unfavorable conditions. Both features differ from firm distress, where executives fix problems with reasonable transparency, often with some contraction and renewed focus on the core business (e.g., John et al., 1992). Observable distress enables workers to self-sort based on risk preferences and demand wage differentials (e.g., Brown and Matsa, 2016; Graham et al., 2019).

One important empirical challenge arises from our research questions; employee data are not commonly available. We use the Longitudinal Employer Household Dynamics (LEHD) and Longitudinal Business Database (LBD) datasets from the U.S. Census Bureau. These are an important data source for addressing questions related to employees in the United States (e.g., Hyatt and McEntarfer, 2012). These data contain workers' entire wage series across employers and a rich set of worker characteristics, such as age, education, gender, and employer location and industry. We combine this employer-employee data with the Securities and Exchange Commission's Accounting and Auditing Enforcement Releases (AAERs) to proxy for fraudulent financial reporting. Our final sample includes about 150 cases of fraud at firms employing a worker in one of 23 states over the period 1989–2008; we use wage data from 362 thousand workers who were employed at these firms in the years leading up to the accounting fraud.[1]

For our main tests, we examine employee wages and turnover during and after fraudulent financial reporting between fraud and control samples. To select the control workers, we propensity score match the fraud firms to control firms within industry and year prior to the

[1] Output from projects that use private U.S. Census data have strict rounding criteria that prevents us from providing a precise observation count in our analyses. In addition, the application process for using U.S. Census data for academic studies requires that individual states approve the project's use of data from that state. For an AAER case to enter our sample, the misreporting firm must have an employee in one of 23 participating states, among other sample criteria. Around 26 participating states is average, (e.g., Walker, 2013; Tate and Yang, 2015, 2016; Dore and Zarutskie, 2017; Goldin et al., 2017; Graham et al., 2019; have 31, 23, 23, 25, 23, and 30 states, respectively).

misreporting. We find that existing employees at fraud firms, compared to the matched sample, have about 4% (9%) lower earnings on average during (after) the fraudulent financial reporting. Discounted cumulative losses are 52% of a worker's annual wage, totaling $275 million for the average firm.[2] Descriptive splits show that worker displacement contributes substantially to these wage effects. As a basis of comparison, cumulative wage losses from the Clean Air Act, competition with China, and bankruptcy are 20%, 23%, and 84%, respectively. (Walker, 2013; Autor et al., 2014; Graham et al., 2019).[3] The separation rate at fraud firms is higher during (after) the fraud period by 4% (18%) on average. After the fraud is revealed, displaced workers are more likely to leave the industry and even the county, taking their next job (if any) elsewhere.

These wage declines exist despite increased employment growth at fraud firms *during* the accounting fraud. Firms shed existing workers—i.e., those employed in the pre-fraud period—replacing them with new, lower paid employees, causing a change to the employee mix. Plausibly, executives engineer this composition change to show headcount growth and keep the wage bill low (e.g., as with cuts to R&D expenditures, McNichols and Stubben, 2008). These results contrast with firms in distress, which lay off workers and do not overbuild (John et al., 1992; Graham et al., 2019). Distressed firms have trouble attracting and are incentivized to lay off newer workers (Brown and Matsa, 2016; Caggese et al., 2019). Ultimately, fraud firms have negative employment growth after the fraud concludes.[4] Afterward, these newly hired employees have cumulative wage

[2] 52.4% cumulative wage losses × $50,340 average wages × 10,440 headcount = $275 million. This equals about 10% of estimated investor losses of $2,819 million ($5,550 total assets × 2.309 Tobin's Q × 22% losses of enterprise value estimated by Dyck et al., 2020, in the 2013 version of their paper).

[3] Walker's (2013) estimates range from 16% to 24% for a 9-year period starting with the implementation of the Clean Air Act. Autor et al.'s (2014) estimates range from 20% to 53% (with interquartile ranges and at the mean using both OLS and 2SLS estimates) for a 16-year period starting in 1992 as Chinese imports to the US increased. Graham et al.'s (2019) estimates range from 38% to 95% for a 7-year period starting in the year of bankruptcy. We compare our cumulative wage loss estimates to the specifications from these papers that most closely match our design.

[4] This result is generally consistent with evidence from Kedia and Philippon (2009) who use employee levels from Compustat. They find greater employee growth during the misstatement period and interpret it as overinvestment

losses equal to 30% of their annual wage. Perhaps this unwinding is discounted by executives who expect to complete the fraud undetected and avoid layoffs, benefiting employees. We perform back-of-the-envelope calculations using the cumulative losses from our main analyses and find that with various assumptions about avoided layoffs, continued growth, and fraud detection rates, frauds are ex ante negative for employees. As an example, avoided layoffs of 5% (e.g., John et al., 1992), continued abnormal growth for 3 years if uncaught, and detection rates of 52% (e.g., Dyck et al., 2020) have a cumulative expected cost for workers of 20% of annual wages.[5]

The results are robust to a variety of specifications, including models with extensive fixed effects; we also use several approaches to control for sample attrition and various control groups. We use employer location and industry data within the LEHD to include specifications with extensive fixed effects to rule out shocks such as regional and industry downturns. To control for sample attrition, we impute missing data as if workers are unemployed, with worker-specific wages, and with a combination of those two using a new, nationwide panel indicating whether a worker has a job anywhere in the U.S. We also vary our control sample in four ways. First, we match firms using hand-collected firm data from the fraud period to control for temporal shocks. Second, we use the employee characteristics data from the LEHD to match subsamples of employees on these characteristics to control for unique worker compositions. Third, we match employees to others in the same industry, without a firm-specific match. Fourth, we match firms

in labor. With the change in employee mix during the fraud, this interpretation is not complete. The departure of existing employees could be a "brain drain" that requires more new employees to perform the same work. Kedia and Philippon (2009) also find higher employee growth *before* the fraud period. For our control firms, we match on pre-fraud employee growth. When using the same matching variables and other design choices from Kedia and Philippon (2009), we replicate their result, and our inferences for the effects on wages and turnover are unaffected.

[5] Continuing this example, fraud detection rates would need to be 15% for workers to have no expected cost from fraud, lower than most estimates from prior papers (e.g., Amiram et al., 2020; Dyck et al., 2020). Moreover, the simulation of the executive's fraud decision in section 5.2 also provides an estimate of employee losses for all fraud—i.e., both caught *and uncaught* fraud—which are also typically negative for employees. We caution that these estimates are sensitive to assumptions, such as the detection rate.

based on fundamentals when the fraud is revealed rather than commenced. These various tests continue to show negative consequences of fraudulent financial reporting for employees.

Finally, we examine the heterogeneous consequences for employees at three different levels to better understand who is affected: at the market, firm, and individual level. First, we separately examine "thin" and "thick" labor markets, i.e., regions with few and many industry-specific employees, respectively. The wage declines are stronger in thin labor markets, indicating that much of the effect likely comes from limited opportunities, consistent with workers job hunting in relatively crowded labor markets (e.g., Moretti, 2011). Second, we show the effects for employees at firms which ultimately go into bankruptcy and not. While the magnitude is larger for the bankruptcy subsample, we continue to find significant wage drops for the non-bankruptcy subsample, i.e., the effect is not isolated to employees of failed firms. Last, we split on employee. We show the effects for leavers and stayers and top 10% and bottom 90% of the pre-fraud wage distribution. Leavers and bottom 90% employees experience negative wage effects from fraudulent financial reporting, unlike stayers and the top 10% of employees who have insignificant cumulative wage losses.

We make several important contributions. First, our paper contributes to an extensive literature documenting other consequences of fraudulent financial reporting (e.g., Erickson et al., 2004; McNichols and Stubben, 2008; Karpoff et al., 2008a, 2008b; Kedia and Philippon, 2009; Dyck et al., 2020). Our analyses add to findings from these papers by measuring the costs of employee turnover and wages at the employee level, estimated at $275 million for the average fraud firm, an additional and previously unmeasured 10% on top of investor losses. Second, we contribute to another extensive literature documenting consequences for employees from a wide variety of shocks to firms (e.g., Gibbons and Katz, 1991; John et al., 1992; Jacobson et al., 1993;

Couch and Placzek, 2010; Walker, 2013; Autor et al., 2014; Hummels et al., 2014; Graham et al., 2019). We show evidence for fraudulent financial reporting, where distinct channels like overbuilding then unwinding and changes to employee mix are relevant to employees. Third, our paper also has policy implications (Leuz, 2018). For example, regulatory reforms intended to reduce burdens associated with mandatory reporting are motivated by job creation, e.g., the Jumpstart Our Business Startups Act (JOBS Act). However, our findings caution that there may be some negative spillovers on workers if eliminating some mandatory information or reducing controls over financial reporting enables more fraud. Another example, the negative effects disproportionately impact workers in thin labor markets and low-paid workers, so policymakers might focus their prevention or damage mitigation efforts due to concerns about rural communities being devastated (e.g., as happened with WorldCom) or inequality.

2. Literature Review and Mechanism Framework

2.1. Literature Review

We summarize three streams of literature: (i) consequences of fraudulent financial reporting, (ii) consequences for employees from shocks to the firm, and (iii) an important subset of (ii): consequences for employees from firm distress.

One stream of literature has examined the broad consequences from accounting fraud for the firm and its peers. For instance, Erickson et al. (2004) show that firms incur real cash outflows to perpetuate fraud; namely, they overpay taxes. McNichols and Stubben (2008) show that firms overinvest in fixed assets, suggestive of internal information frictions. Kedia and Philippon (2009) show some effects related to ours with aggregated employee count and GAO restatement data. Kedia and Philippon (2009) also show overinvestment, consistent with McNichols and Stubben

(2008), and have some evidence on increases in productivity after restatements. Beatty et al. (2013) investigate the spillover effects from high-profile accounting frauds on peers' investment, which increases during the fraud period and could be facilitated by equity analysts. Li (2016) builds on Beatty et al. (2013) to show that these peer spillovers are more general, not confined to high-profile fraud and observable in many types of investment, e.g., R&D. Other papers document broader cost estimates; as mentioned in the introduction, Karpoff et al. (2008b) find that firms lose about 29% of equity value. Of the measured decline, only 12% is expected legal penalties while 88% is a reputational penalty. Dyck et al. (2020) find that firms lose about 22% of enterprise value.

Academics also have documented the effects of fraudulent financial reporting for directors and executives, highly visible employees at the top of the firm. Early evidence from Agrawal et al. (1999) and Beneish (1999) suggested that firms suspected or charged with fraud did not have unusually high turnover among executives. Subsequent papers have documented significant career consequences for directors and executives. For restatements (not always fraud), Srinivasan (2005) finds in a three-year window that audit committee director turnover is 48% for firms that restate earnings downward and 33% for a performance-matched sample. Desai et al. (2006) find that 60% of restating firms turnover at least one top manager in a two-year window compared to 35% for matched firms. Karpoff et al. (2008a) examine SEC and Department of Justice enforcement actions and find that 93% of identified executives lose their job, with some facing criminal charges and penalties that include jail time. In contrast to these papers, our focus is on all employees and the heterogeneous effects of accounting fraud for different groups of employees.

Prominent papers in labor economics document costs to employees and cross-sectional variation of mass layoffs. For example, Gibbons and Katz (1991) present a theoretical model of layoffs: when firms can choose whom to lay off, the demand-side of the labor market infers that

laid-off workers are of low ability, i.e., "lemons." They show evidence consistent with this among laid-off white-collar workers who experience about a 6%-9% wage decline, depending on specification. In their seminal paper, Jacobson et al. (1993) use administrative data from Pennsylvania to observe workers' wages in series; they find that high-tenure workers who separate from distressed firms suffer immediate (long-term) losses averaging 40% (25%) per year. Importantly for our analyses, they find that displaced workers' losses depend primarily on local labor-market conditions but not other worker attributes. More recently, Couch and Placzek (2010) revisit Jacobson et al.'s (1993) result due to concerns about magnitude estimates from a single, primarily industrial state during the 1970s and 80s. Couch and Placzek (2010) use administrative data from Connecticut and again find meaningful wage losses for displaced workers. They find that shortly after the separation, wages for workers displaced through mass layoff drop immediately (over the long-term, i.e., 6 years) 32% (12%) percent.

More recently, Walker (2013), Autor et al. (2014), and Hummels et al. (2014) examine the impact to employees from other shocks to firms, including environmental regulation, globalization, and offshoring, respectively. Walker (2013) finds workers affected by the 1990 Clean Air Act lose, cumulatively, about 20% of their earnings on average. Autor et al. (2014) find that over 1992-2007, U.S. workers exposed to trade competition with China lose, again cumulatively, 46% of their earnings when moving across the inner quartile range and 23% at mean trade exposure levels. Finally, Hummels et al. (2014) examine the impact of offshoring on Danish workers' earnings; the authors find that, as an estimated size, when a firm doubles its offshoring, unskilled workers can expect cumulative wage losses of about 12%.

One important stream of literature examines the labor market impact of firm distress. An earlier paper that focuses on product market (rather than financial) distress and conditions the

sample on recovery, John et al. (1992) show that firms very quickly shed about 5%-6% of their employees using levels data from Compustat. Using similar Compustat data, Whitaker (1999) shows that firms which take remedial actions during distress, like cutting employees, tend to recover. These sample firms publicly announce that they are taking actions to recover from their poor performance, including reducing headcount.[6] Falato and Liang (2016) find that debt covenant violations are associated with 10% drops in headcount. An important benchmark for our paper is Graham et al. (2019), who also use the U.S. Census LEHD data. They find that workers at severely firms experience cumulative wage declines of 84% over seven years due to bankruptcy, and employee attrition increases 11% compared to a matched sample. Finally, other papers provide some relevant facts for the interpretation of our results. For example, Brown and Matsa (2016) show that job seekers understand and respond to firms' financial conditions. As an employer's distress increases, it receives fewer and lower quality applicants for posted jobs.[7] Building on these labor economics and finance papers, our paper documents that excessive optimism or unsuccessfully hiding bad performance backfires on the firms' employees due to building up the workforce under information asymmetry and subsequent labor market disruptions from unwinding that buildup.

[6] In a working paper, using Swedish employer-employee data, Baghai et al. (2020) show that firms do worse retaining talented employees when distressed. Financial distress can be costly to the firm because employees value the insurance-like features that a stable job can provided (Baily, 1974). Davis et al. (2014) provide a benchmark from private equity investments. Private equity investments are associated with 6% cumulative job losses over two years, then job gains.

[7] For other examples, workers are also responding to accounting specific information provided by firms (e.g., Choi et al., 2020; deHaan et al., 2020). And relevant for our tests on heterogeneity, spillover effects from bankruptcy are particularly severe in thin labor markets (Bernstein et al., 2018).

2.2. A Framework for the Impact of Fraud on Labor Markets

In this subsection, we discuss two features of accounting fraud and associate these features with mechanisms that could affect workers, specifically wages and turnover. We also describe how these features differ for firms in distress.

Overemployment: Hiring and Turnover Decisions

Executives in accounting-fraud firms appear to overinvest in physical capital and may also over-hire employees to bolster the perception of the firm (McNichols and Stubben, 2008; Kedia and Philippon, 2009). In many cases, Schrand and Zechman (2012) find that executives take actions which show growth and end up committing fraud because they hope it will go uncaught while they attempt to correct earlier, excessive optimism. However, they are still carefully managing reported performance. McNichols and Stubben (2008) show that capital expenditures are abnormally high while R&D is not; the authors suggest that R&D causes an immediate drag on profitability so is a less appealing investment strategy to propagate fraud (and perhaps complete it undetected). Likewise, executives could grow the number of employees during accounting fraud (e.g., like periods of accounting misstatements, Kedia and Philippon, 2009). However, executives simulating employee growth would increase wages, lowering reported performance, i.e., increased wage expenses reduce current profits. Executives could instead shed existing employees and hire even more new employees, replacing high paid with low paid workers and so change the employee mix to have more, lower paid, shorter-tenured employees.

Conversely, when firms are distressed, executives take a different approach with respect to employees. If the executive quickly acknowledges problems and develops a plan to restructure or refocus the business around its core operations, some employees might be laid off. However, these quick layoffs could contribute to a successful recovery (e.g., John et al. 1992; Whitaker, 1999).

Further, distressed and constrained firms do not necessarily substitute long-tenured workers with new or short-tenured workers. Distressed firms could have trouble attracting new workers (Brown and Matsa, 2016) or be forced to pay wage premiums (Graham et al., 2019). Constrained firms also shed newer workers if firing costs for long-tenured workers or training costs are high (Caggese et al., 2019).

Information Asymmetry

A preeminent feature of fraudulent financial reporting is that executives are falsifying public information about the firm, which often shows better performance than the hidden and unmanipulated information. Workers use financial information from the firm (Choi et al., 2020; deHaan et al., 2020). When workers keep or take a job in the presence of these informational asymmetries, they are misled about the likelihood of suffering a negative shock in the future. If the firm does not improve under cover of the fraud and the fraud is revealed, employees only then learn that the firm has worse prospects compared to what had been falsely reported.

This misinformation combined with the overemployment described above would affect workers through three mechanisms. First, employees have risk preferences with respect to jobs, including distress and information quality (e.g., Brown and Matsa, 2016; Choi et al., 2019). As a result of the misreporting, they may take a job or keep the job at the fraud firm despite other options. Second, when employees work for a firm, they accumulate firm- (and industry-) specific human capital (Becker 1993). Because specific capital loses value when the worker is displaced, there may be heightened but unobserved turnover risk which would distort the worker's investments in general versus specific capital. Third, due to overinvestment at the fraud firm (or even in the fraud firm's industry, e.g., Beatty et al., 2013), workers with similar skills are likely to lose jobs at the same time. Workers will be searching for their next job in an unfavorable local

labor market condition: the labor market will be "crowded," i.e., many, similar workers will be looking for a job at the same time. Unwinding the overemployment would cause displacement. And these mechanisms, conditional on displacement, would be costly to workers in terms of wages.

Again, with a distressed firm, outcomes are likely to be different. With accurate information about less optimistic or poor performance at the firm, employees might voluntarily switch jobs or applicants might take a different job elsewhere (Brown and Matsa, 2016). If an employee continues to work for the firm, she may demand a wage differential for job risks associated with distress (Graham et al., 2019). Finally, the executives could take remediating actions early, preventing worse outcome later (e.g., Whitaker, 1999) and alert workers early such that potential job transitions are more successful, e.g., shorter and with lower wage drops (e.g., Malik, 2020).

3. Data and Research Design

3.1. Accounting and Auditing Enforcement Releases

Our sample for fraudulent financial reporting is the enforcement actions taken by the Securities and Exchange Commission (SEC), Accounting and Auditing Enforcement Releases (AAERs). This sample identifies cases of accounting problems that can be connected with prosecutable, fraudulent behavior by executives (Schrand and Zechman, 2012). We use UC Berkeley CFRM's dataset. We do not screen for different types of fraud, except that we require annual financial statements to be fraudulently misstated (i.e., we only screen cases with affected quarterly financial statements).[8] We want our results to show average effects for any serious fraud that rises to the level of SEC enforcement and be as general as feasible given other sample

[8] The CFRM database also excludes cases of simple bribes or disclosure violations without earnings misstatements.

limitations. Many prior papers have used these enforcement actions across a range of topics, for instance, to estimate, describe, and measure effects of fraudulent financial reporting (e.g., Feroz et al., 1991; Beneish, 1999; Farber, 2005; Dechow et al., 2011; Groysberg et al., 2017).

The AAER sample involves a tradeoff where Type I errors for identified misreporting are very low but sample size tends to be small and spread out over many years (Dechow et al., 2010). Karpoff et al. (2017) indicate that AAER data perform relatively well across a variety of metrics, such as capturing fraud cases from Karpoff et al.'s (2017) Section 13(b) financial misrepresentation data or low omissions of same types of fraud (i.e., see their Table 8). Karpoff et al.'s primary criticism about the AAER data is that the release date—the specific date when the SEC publishes the release—is not when outsiders (e.g., stock market participants) learn about the misreporting. Karpoff et al. (2017) find that "trigger events" occur 1.6 months after the misreporting period for the median case. Our analyses focus on overbuilding and unwinding having effects in labor market outcomes and are agnostic about capital market outcomes. We test whether unwinding begins in the year after the misreporting concludes (e.g., a "post" fraud period) and assume that misreporting is revealed to the public in that year as well. This assumption is consistent with the median trigger event date from Karpoff et al. (2017).[9]

3.2. U.S. Census data

We combine this AAER data with worker-firm matched data from the U.S. Census Bureau Longitudinal Employer-Household Dynamics (LEHD) and Longitudinal Business Database (LBD) data.

[9] In untabulated analyses, we match the trigger event dates from Call et al., (2018) by firm and misreporting period to AAERs. We confirm that about 78% have trigger event dates that occur within a year of the last misreported annual financial statements.

The LEHD data have a comprehensive coverage of workers, on average covering 96% of all private-sector jobs across years (e.g., Abowd et al., 2005). We have data from 23 states participating in the LEHD program. These data include wage data when the earnings are covered by a state's unemployment insurance program and generally include salaries, bonuses, equity, tips, and other perquisites (e.g., meals, housing, and retirement contributions, among others) (BLS, 2016). We observe these earnings as quarterly and annual pay. Self-employed, unemployed, and workers who move to non-participating states are not observable in the LEHD data.[10] The data allow us to track the wages of workers who were employed at accounting-fraud firms but have since moved to other firms. We also use the individual characteristics provided by the LEHD data to separate the effects of misreporting and employee characteristics (e.g., gender, education, and experience) on wages. We require that employees be between 22 and 50 years old in the year prior to the fraud period, consistent with other papers (e.g., Davis and von Wachter, 2011; Autor et al., 2014); this requirement generally limits the sample to workers who are (or desire to be) full-time participants in the workforce.[11] We also require that the worker's annual real wages are higher than $10,000 to exclude temporary workers.

The LBD data contain aggregated, establishment-level information (e.g., Davis et al., 2014; Giroud and Mueller, 2017). It covers the universe of non-farm industries across the United States so is not limited to the 23 participating states like the LEHD data. The LBD data come from the

[10] If a worker has multiple employers within a year that are observable, we aggregate wages across employers and use the highest wage as the employer match for the year. We conduct several tests for missing worker wages and sample attrition. See section 4.3.

[11] We generally follow prior papers in selecting this age range (e.g., Davis and von Wachter, 2011; Graham et al., 2019). With the pre period window, the youngest a worker could be is 19 years old. With the average length of a fraud, 2 years, the oldest a worker would be is 58. Results are robust to using other age ranges, such as employees between 20 and 55 the year before the fraud period.

IRS and include variables such as wage bill and employment. We use these data to track employee growth within a misreporting firm over pre-fraud, fraud, and post-fraud periods.[12]

To publicly report our results, Census policy requires that we test for the influence of large firms (or in our case, large firms and large frauds). The findings of these tests indicate that our results do not reflect a handful of influential firms nor frauds. We also are restricted from mentioning any firm in our sample, so we cannot speculate which large frauds might give cause for concern. However, we are confident that this process reasonably eliminates the possibility that one or a handful of frauds drive the results.

3.3. Research design and matching

Our research design allows measurement of effects from fraud to be dynamic over the misreporting's lifecycle. We treat the misreporting as having three distinct periods. (i) "Pre" is the three-year period prior to the beginning of the fraudulent misreporting. (ii) "Fraud" is the period identified by the AAER, normalized to a maximum of three years. And (iii) "Post" is the six-year period after the fraud. This approach follows McNichols and Stubben (2008), though we stack, rather than truncate later years (more than three years) of the fraud.[13] For most analyses, we examine existing employees, i.e., those employed in the pre-fraud period; though, we also use a

[12] The Compustat-SSEL Bridge (CSB) (covering 1981-2005) and the Standard Statistical Establishment List (SSEL) (covering later years) use primarily CUSIPs to link Compustat to LBD. We supplement these links by matching Employer Identification Numbers and company name, address, and industry in both data. We merge the Computstat-LBD data with the LEHD files using the Employer Characteristics Files (ECF). These linking files are widely used in prior literature (e.g., Giroud and Mueller, 2017; Graham et al., 2019). Finally, we merge with CFRM using CIKs (current and historical).

[13] Workers at the fraud and matched control firm have identical treatment with the event-time indicators (e.g., *Fraud(t)*, *Fraud(t+1)*, *Fraud(t+2)*, *Post(t+3)*, etc.). A long-lasting fraud (four or more years) would have multiple calendar years where *Fraud(t+2)* equals one. For example, a seven-year fraud would have five calendar years coded as *Fraud(t+2)*. *Post(t+3)* only equals one in the first year after the fraud ends, no matter its length. For one-year, two-year, and long-lasting frauds, t+X does not match sequential calendar years. In the Internet Appendix Figure 1, we show effects separately for one-, two-, and three & more-year frauds. Because of disclosure restrictions, we are unable to examine additional subsamples of long-lasting frauds.

sample of new employees, i.e., those hired during the fraud period, to show cross-sectional effects and information frictions. We match the fiscal year of the firm to the calendar year of the LEHD (which only measures data in calendar quarters).

We create a matched sample of fraud and non-fraud firms to control for firm fundamentals. We require that potential control firms be covered by the LEHD data (i.e., these firms will have at least one employee hired in one of the 23 states for the pre fraud and fraud periods). We perform a propensity score match within industry-year, using 3-digit SIC industry codes from the firm-year prior to the misreporting.[14] We follow McNichols and Stubben (2008) and Schrand and Zechman (2012) and match on characteristics the year prior to the fraud. Further, because executives already take real (and different) actions in the first year of the fraud, characteristics in the final year before deciding to commit fraud is our preferred year for a match. We estimate the following cross-sectional probit model on the CFRM-Compustat-LBD-LEHD sample to obtain firm-year scores to match fraud to non-fraud firms:

$$\begin{aligned} \textit{Fraud-Firm Indicator}_{i,t-1} = {} & \beta_0 + \beta_1 \times \textit{Size}_{i,t-1} + \beta_2 \times \textit{Sales Growth}_{i,t-1} + \beta_3 \times \textit{Return on} \\ & \textit{Assets}_{i,t-1} + \beta_4 \times \textit{Leverage}_{i,t-1} + \beta_5 \times \textit{Tobin's Q}_{i,t-1} + \beta_6 \times \textit{Employee Growth}_{i,t-1} + \\ & \beta_7 \times \text{Ln}(\textit{Avg. Annual Real Wages}_{i,t-1}) + \varepsilon_{i,t-1}. \end{aligned} \quad (1)$$

Appendix Table A contains variable definitions. Equation (1) indexes firm with *i* and fraud event-time with *t*. In Appendix Table B, we report the results of the probit model. Consistent with prior literature, *Size* and *Tobin's Q* significantly and positively correlate with *Fraud-Firm Indicator* (e.g., Farber, 2005; Schrand and Zechman, 2012). *Sales Growth* and *Return on Assets* also positively correlate with *Fraud-Firm Indicator*. We also include *Leverage*, *Employee Growth*, and the log of *Avg Annual Real Wages* in the model because prior papers indicate that these

[14] If a good match is unavailable in the 3-digit SIC industry, we pick a match from the 2-digit SIC industry (following Graham et al., 2019). Results are qualitatively similar when using only 2-digit SIC industry codes.

variables are correlated with employee wages and employee composition (Berk et al., 2010; Chemmaunur et al., 2013; Ouimet and Zarutskie, 2014).

Our main empirical tests measure wage effects for existing employees from the fraud and control firms in the matched sample.[15] We estimate the following OLS specification characterizing workers' wages:

$$\begin{aligned}\text{Ln}(\textit{Annual Real Wages}_{j,\tau}) = \alpha + \\ \beta_{1,p} \times \sum_{p=1,2,3} \textit{Pre(t-p)}_{j,\tau} + \beta_{2,p} \times \sum_{p=0,1,2} \textit{Fraud(t+p)}_{j,\tau} + \beta_{3,p} \times \sum_{p=3,4,5,6,7,8} \textit{Post(t+p)}_{j,\tau} + \\ \beta_{4,p} \times \textit{Fraud Ind.}_{j} \times \sum_{p=1,2,3} \textit{Pre(t-p)}_{j,\tau} + \\ \beta_{5,p} \times \textit{Fraud Ind.}_{j} \times \sum_{p=0,1,2} \textit{Fraud(t+p)}_{j,\tau} + \\ \beta_{6,p} \times \textit{Fraud Ind.}_{j} \times \sum_{p=3,4,5,6,7,8} \textit{Post(t+p)}_{j,\tau} + \\ \sum \beta_m \text{ Worker Controls}_{j,\tau} + \sum \beta_k \text{ Fixed Effects}_{j,\tau} + \varepsilon_{j,\tau.} \end{aligned} \quad (2)$$

Equation (2) indexes worker with j and calendar year with τ. Fraud periods vary in calendar time depending on the worker. Worker controls include interactions of *Female Indicator*, *Education*, and *Experience* (e.g., Topel, 1991).[16] In all specifications, we include worker and year fixed effects. In some specifications, we interact industry and county fixed effects with the year effects to rule out shocks such as regional and industry downturns. The period indicators (i.e., *Pre*, *Fraud*, and *Post*) span the sample except the baseline year, the year prior to *Pre(t-3)*. Moreover,

[15] Wages are scaled using the CPI to 2010 price levels.

[16] The *Gender* and *Education* main effects are collinear with the worker fixed effects. *Experience* is also collinear with the main effects; that is, when *Experience* is demeaned by worker, it is effectively equivalent to a sequential count of the number of years in our sample and thus a combination of the worker fixed effect and calendar year indicators. The interactions are included in the regressions, but we do not report the estimates for parsimony. We find negative coefficients on *Female* × *Experience*, consistent with greater wage increases for men with additional experience compared with women (Bertrand et al., 2010). And we find positive coefficients on *Experience* × *Education*, consistent with greater wage increases with experience for the more highly educated (e.g., workers with MBAs or JDs have steeper wage trajectories over their careers than high school dropouts) (Heckman et al., 2003).

we include fixed effects that take into account both calendar year (β_k) and event year effects (β_1, β_2, and β_3).[17] This approach generally follows Graham et al. (2019), except for the consideration of a fraud period. We provide a detailed timeline in Figure 1 that maps out these period indicators.

This specification is a difference-in-differences approach to estimate the effects of fraudulent financial reporting. β_4 is estimated wages for workers at fraud firms incremental to those at control firms prior to the misreporting. If the matches are reasonably well chosen, we expect the estimated coefficient to be insignificantly different from zero and not exhibit any pre-fraud period trends. β_5 measures the incremental wages of fraud-firm employees for the fraud period. This measure is our first coefficient of interest; we infer the consequences for employees *during* the fraud from this coefficient estimate. β_6 measures the incremental wages for employees of fraud firms during the post-fraud period. This measure is our second coefficient of interest; we infer the consequences for employees *after* the fraud from the coefficient estimate. Our assumption for both coefficients is that wages would have evolved (in the absence of fraudulent financial reporting) for employees of AAER firms during and after the fraud as wages have evolved for control-firm employees. Moreover, while we cannot exogenously impose fraud, we control for firm fundamentals and the distribution of firm performance draws prior to the fraud period.

Besides these dynamic wage effects, we also use these coefficient estimates, β_5 and β_6, to calculate the present value of cumulative earnings losses discounted to the year before the fraud (i.e., the end of *Pre(t-1)*) and the final year of the fraud (i.e., the end of *Fraud(t+2)*). We convert the coefficients to percent wage changes relative to the baseline year and use an inflation-adjusted

[17] Other cross-sectional fixed effects (e.g., fraud and control firm, cohort, etc.) are collinear with worker fixed effects so are not included in equation (2).

discount rate, following prior literature (Graham et al., 2019). Next, we examine attrition rates for existing employees at fraud and controls firms using the employee-level data.

4. Main analyses

4.1. Sample description

Table 1 Panel A provides comparisons of our matched fraud and non-fraud (control) firms. In total, our sample contains about 150 fraud and 150 control firms. Our matching process described in section 3.3 generates a reasonably well-balanced sample, with only some difference in *Assets* between the two groups; this difference is perhaps unsurprising because we do not match firms using *Assets*. We do not find significant differences between fraud and control firms when comparing *Size*, *Sales Growth*, *Return on Assets*, *Leverage*, *Tobin's Q*, *Employee Growth*, *Avg Annual Real Wages*, and *Employees*.[18] The average, firm-wide annual wages are comparable for fraud and matched control firms and equal to about $54 or $55 thousand normalized to 2010 CPI price levels. As a comparison, per capita income in the U.S. in 2019 was $33 thousand using the Census's Current Population Survey (again at 2010 price levels). Table 1 Panel B gives descriptive statistics of firm characteristics for fraud firms with LEHD data, i.e., our sample, and all fraud firms with Compustat data. Firms with employees in more states have a higher likelihood of entering the LEHD data, so we expect our sample to contain larger and more mature firms. This is consistent with the differences from Table 1 Panel B. Specifically, our sample fraud firms are larger, more profitable, have lower leverage, and have lower growth prospects.

[18] Other pre-fraud period comparisons of the fraud and control firms indicate that the matches are good. We test differences in all firm characteristics in periods *Pre(t-4)*, *Pre(t-3)*, and *Pre(t-2)*; the characteristics between fraud and control firms are not different from each other in 18 out of 21 tests as shown in Internet Appendix Table 2. Also, we can match on other subsamples of these variables and qualitatively our findings are unaffected.

Table 2 presents descriptive statistics on the individual characteristics of employees of fraud and control firms. This sample includes existing employees who work for the fraud or control firm in the two years prior to the fraud period, that is, *Pre(t-2)* and *Pre(t-1)*. These data (and calculated differences) are from the last year of the pre-fraud period, *Pre(t-1)*. At fraud firms, employees have similar education and gender. The annual real wage for individual workers in our sample is equal to about $71 thousand at fraud firms and $54 thousand at control firms.[19] Employees at fraud firms are older by a year and, consequently, have slightly more experience.[20]

4.2. Results for wages and displacement

Table 3 contains our main result, including estimates of dynamic wage effects during and after fraudulent financial reporting and calculations for the wage effect as present values before the fraud and before the fraud's revelation. Across columns, we increase the number of fixed effects; in columns 1, 2, and 3, we estimate models with worker effects and year effects, year-industry, and year-industry-county effects, respectively. For the dynamic wage effects, we find consistently negative wage coefficients in the post-fraud periods for employees who work(ed) at fraud firms. However, wage drops during the fraud period are negative but not significant at conventional levels. The magnitudes for the post-fraud period are meaningfully negative, ranging from -5% to -13%. The worst years for workers of fraud firms are *Post(t+4)* through *Post(t+7)*, bottoming out then recovering somewhat, depending on the specification. To get a better sense of the trends, we depict columns 1 and 3 graphically in Figure 2 with confidence interval estimates.

[19] These employees have wages greater than the firm-wide averages. Our main sample focuses on existing employees with two years of work experience at the firm. The firm-wide averages include all employees in the LBD data, including temporary workers and those with shorter tenure, who naturally have lower wages (Topel, 1991).

[20] In a separate test, we match employees at fraud firms with employees at non-fraud firms using individual characteristics including, e.g., age and education. See section 4.3.

We also examine evidence for common trends using the first three coefficient estimates. In columns 1 and 3, we observe that employees in the pre-fraud period have similar wage changes compared with workers at control firms. In column 2, *Pre(t-3)* and *Pre(t-1)* are negative and significant in the pre-fraud-period, indicating potentially some negative wage trend.[21] Otherwise, the estimated coefficients for the pre-fraud period are not significant (though negative). Overall, these tests indicate that the year and year-industry-county effects remove much of the variation from local shocks that could pre-date the fraud. When controlling for these explanations, the onset of negative wage effects are relatively sharp and start after the fraud is revealed.

The final two rows of Table 3 give present value calculations for cumulative, estimated losses as a percent of workers' annual wages. For these calculations, we convert the coefficient estimates to percent wage losses for each year and discount (using 4.47%, the inflation-adjusted average corporate BBB bond yield during our sample period) to the end of the pre-fraud period and the end of the fraud period.[22] Workers experience cumulative wage losses of about 52.4% of their annual wages measured at the beginning of the fraud and 47.5% measured at the beginning of the post fraud period, using estimates from column 1. These cumulative wage losses are greater than economic shocks such as the firms offshoring work (12%), being exposed to certain regulations (20%), and competition with China (23%) (Hummels et al., 2014; Walker, 2013; Autor et al., 2014), but smaller than economic shocks such as firm bankruptcy (84%) (Graham et al., 2019).

[21] In most aspects of the research design, we follow Graham et al. (2019); however, we use specifications with worker and year fixed effects (instead of worker and year-industry effects like Graham et al., 2019) as our main specification for interpretation and subsequent heterogeneity tests. Thus, we avoid having these significant pre period trends contribute to the present value calculations. Adding dense fixed effects limits remaining variation to measure outcomes of interest because we require some fraud and some control observations in same county, same industry, and same year. This can cause measurement issues when limiting the sample, e.g., for cross-sectional splits.

[22] This discounted value approach and discount rates are similar to those in other papers (e.g., Sullivan and von Wachter, 2009; Davis and von Wachter, 2011; Graham et al., 2019). Corporate yield data come from the Bank of America and Moody's series maintained by the St Louis Federal Reserve Bank. We use the variance-covariance matrix from the coefficient estimates and apply the delta method to calculate t-statistics for these present values.

Columns 2 and 3 show somewhat larger and smaller estimates, respectively. The magnitudes are economically meaningful and primarily driven by wage losses after the fraud is revealed.

In Table 4, we examine whether employees of fraud firms are more likely to leave a firm, an industry, and a county during or after fraud periods. We generate dummy variables that indicate whether an employee stays working (i) at the firm, (ii) in the industry, or (iii) in the county for all years in event time during and after the fraud. For industry and county, we indicate with the industry and location of the employee's *next* job.[23] We present the averages for these dummies for employees of fraud firms in columns 1 and control firms in column 2. Initial employee attrition rates from the firm, industry, and county for employees who leave the control firms are 13%, 12%, and 13%, respectively, (e.g., 100% minus 87.3%). The existing employees of fraud firms are more likely to leave the firm by 5% and leave the industry by 6.7% in the first year of the fraud. Fraud firm employees do not incrementally leave the county for their next job during the fraud. In Panel B, during the post fraud period, these differences increase substantially. After the fraud, employees are significantly more likely to leave the fraud firm, the industry, and the county for their next job.

4.3. Sample Attrition and Different Matching Approaches

As discussed above, our data is limited to 23 states and is subject to sample attrition associated with worker movements to states that are not in our sample. This could cause measurement problems of wage effects when workers differentially move away from fraud and control firms.

[23] If the worker has a subsequent, missing observation, we consider them to have left the firm, industry, and county due to data limitations. The "stay" county-level measure is biased downward if the worker stays unemployed in the same county. Alternatively, it could be that the worker leaves the county to a state that is not in our sample in order to stay at the same firm or in the same industry. So, we may underestimate "stays" for firm and industry. There is another subtlety with respect to industry and county measurement. Firms have establishments that have different industry coding and in different counties. Workers can stay at the same firm but switch industry or switch county or switch both. This could cause the percentage of workers staying in the industry or staying in the county to be lower than the percentage staying at the firm.

Moreover, when the wage data is missing, we do not infer zero wages. This measurement choice potentially underestimates the costs of some job switches because we do not include the zeros for workers with long unemployment spells. Other papers using LEHD data also face this issue (e.g., Walker, 2013; Graham et al., 2019). We follow these papers and impute missing earnings in the LEHD data using two separate approaches. We also introduce a new imputation approach with nationwide data that is not limited to the specific states that enter our main sample.

Table 5 Panels A through C present results with the various imputation approaches to mitigate the effects of sample attrition; these panels give the present value calculations. For all imputations, we use the most recently observable industry and county when a worker's wage data is missing. In Panel A, for both fraud and control employee groups, we replace missing earnings with the 10th percentile from our sample's wage distribution. This imputation treats the workers who drop from our data as effectively unemployed and earning some minimal wage. In Panel B, missing annual earnings are replaced with the worker's last observed quarterly earnings multiplied by four. This imputation assumes that those who drop from our sample are moving to work in another state (where we lack access to their wage information) or become self-employed and have similar annualized earnings.

Finally, in Panel C, we use newly available, nationwide data that indicate when a worker is employed *anywhere*. With these indicators, we replace missing wages with the 10th percentile (i.e., as in Panel A) when a worker is not employed anywhere and replace missing wages with a worker's most recently observed annualized wage (i.e., as in Panel B) when they are employed somewhere. We prefer this approach because it uses unique data to differentiate between those who are likely to be unemployed and those who likely move out of state and continue working. Across these approaches, magnitudes and inferences are highly similar, except with some

attenuation in Panels A and B when using year-industry-county fixed effects.[24] In particular, our preferred approach is to use the nationwide data; these results have nearly equivalent magnitudes and significance as our main results.

For our next tests, we vary the control samples in four ways and present these results in Table 5 Panel D. First, we match fraud firms using hand-collected data from the fraud period to control for potential contemporaneous shocks.[25] These results are shown in column 1 and are quantitatively similar to the main sample's results. Second, we use the employee characteristics data from the LEHD to match subsamples of employees from fraud and control firms. We one-to-one match employees on age, education, experience, gender, and pre-fraud wage decile without replacement. In performing this subsample match, we lose 7 million employee-year observations (80% of our main sample). The findings are shown in column 2. While all negative, only coefficients for periods *Post(t+4)* and *Post(t+5)* are significant at conventional levels indicating that like employees suffer negative wage consequences.[26] However, employee composition at fraud and matched control firms plays some role in wage trajectories; for example, coefficient

[24] A reasonable assumption is that missing observations biases the coefficients towards zero and imputation of 10th percentile wages would result in greater estimated losses from fraud. However, workers from both fraud and control firms drop from the sample due to unemployment *and* other reasons (e.g., switch states), so imputations could cause measurement error or, worse, systematically misassign 10th percentile wages to workers taking similar jobs elsewhere. So, attenuation from 10th percentile imputation is possible and can be explained. Panel C, where the result does not attenuate with the nationwide data, confirms this explanation.

[25] We separate the fraud sample into revenue misreporting and non-revenue misreporting. For the revenue misreporting subsample, we gather unmanaged sales data from, in order: (i) differences between Compustat-Snapshot "As First Reported - Annual" and "Most Recently Restated - Annual", (ii) AAER reported annual misstatement amounts, (iii) restatements on SEC EDGAR database, and (iv) a Factiva and Google search for archival news documents reporting on the fraud. We use this hand-collected data to construct a *Sales Growth* variable measured from *Pre(t-1)* to *Fraud(t)* and include this variable in our propensity-score-matching model along with the other variables noted in equation (1). Matching model and descriptive statistics are presented in Internet Appendix Table 4.

[26] The estimates for the present values for this employee composition matched sample, i.e., -23% for pre fraud and -21% for pre revelation, are similar to cumulative wage loss estimates for workers negatively affected by the Clean Air Act of -20% (Walker, 2013) and trade competition with China of -23% (Autor et al., 2014). This matching approach is restrictive and is not used by other papers measuring cumulative wage losses (e.g., Walker; 2013; Autor et al., 2014; Graham et al., 2019). Instead, these papers use Mincer earnings regressions, controlling for worker characteristics and using control groups from similar industries as we have done in our main design in Table 3.

magnitudes in periods *Post(t+6)* through *Post(t+8)* are closer to zero, indicating faster reversion to "baseline" wages in the post fraud period than other specifications. Third, we match fraud firm employees to random employees within industry. With this alternative control sample in column 3, results are comparable to the main result but also indicate negative wage effects during the fraud for workers at fraud firms. That is, workers at fraud firms fare quite poorly relative to the average worker in their industry.

Fourth and finally, we match fraud firms to control firms during the fraud's revelation (i.e., *Post(t+3)*). A natural, alternative set of control firms are those with ex post characteristics of firms that cannot resolve business problems under cover of fraud and are discovered to have committed fraud. This final matched sample tests whether wage outcomes for employees at fraud firms compared to employees at control firms that have ex post outcomes that are like discovered fraud firms. Because fraud firms may be unwinding some fraud-related overbuilding of employees (more on the mechanism is section 5), matched controls will similarly have characteristics like sharply reducing headcount, etc. Thus, matching on a similar outcome for a non-fraud firm generates a control group that could also be unwinding excessive growth and control for some worker consequences from fraud measured with matching in the pre fraud period. So, the effects could be muted with this alternative control sample but still capture mechanisms such as job search or information frictions. In column (4), we continue to find negative wage consequences during the fraud (i.e., *Fraud(t+1)* and *Fraud(t+2)*) and after (i.e., *Post(t+5)*). Other years have negative coefficients but are not significant at conventional levels. We find cumulative wage losses of 44%, similar to the 52% from Table 3, though are only significant at one-sided p-values less than 10%.[27]

[27] Related to fraud firms sharply reducing headcount, perhaps many frauds are revealed during recessions, like the financial crisis, and so negative wage effects could be driven by tough labor market conditions from these macro

5. Mechanism and Heterogeneity

5.1. Mechanism: Overbuilding, Employee Mix, and Information Friction

Above, we argue that managers overbuild in conjunction with, or as part of, the accounting fraud. This argument is consistent with the findings from several prior papers (e.g., McNichols and Stubben, 2008; Kedia and Philippon, 2009; Schrand and Zechman, 2012). We show the dynamics of employee growth during the pre-fraud and fraud periods. In Figure 3, we present the trend of fraud firms' employment decisions measured as year-on-year employee growth. We include growth at control firms and for the industry average as comparisons. Before the fraud, fraud firms have high, positive employee growth ranging between 14% and 16% annually (control firms range between 11% and 15%; industry average growth rates are between 7% and 8%). During the fraud, employee growth rates remain quite high; fraud firms' rates average 9%, controls average 3% and the industry averages 5%. These percentages indicate that the fraud firms were hiring during the fraud period and hiring at rates much greater than these comparison samples.

This finding of higher total employee growth points out the subtlety of the findings in Table 4, where existing employees are displaced more frequently than existing employees at control firms. These two findings suggest that fraud firms are building up their workforce but substituting new employees for long-tenured employees, changing the employee mix, a result which is inconsistent with distressed firms. Distressed firms are shedding workers; for example, Graham et

trends. We prefer to retain all frauds in our sample because the average labor market condition when fraud is revealed matters for estimating the average effects for employees from fraudulent financial reporting. However, in Internet Appendix Table 5, we conduct a test that excludes frauds revealed in years 2000, 2001, 2007, 2008, and 2009 and find some attenuation in present value calculations. This is consistent with more negative effects for employees when fraud is revealed during recessionary periods. The magnitude excluding these years is still large, with estimated cumulative wage losses of 30% to 36%, larger than estimates from, e.g., trade competition from China of 23% (Autor et al., 2014). On average, 28% of all AAER fraud cases are revealed during these recession years. These cases are worse for employees and are highly relevant for our estimates.

al. (2019) show that headcount drops by about 6% annually in the two years leading up to bankruptcy.[28] Also, these distressed firms churn new employees and keep long-tenured employees, giving up growth from the new workers and squeezing productivity out of the experienced ones (Caggese et al., 2019). For fraud on the other hand, if existing employees are more expensive than newly hired employees, fraud firms' executives plausibly engineer this composition change to show headcount growth and keep the wage bill low.[29]

In Table 6, we test employee mix and new versus leaving employee wages directly. Panel A shows that before and during the fraud period, fraud firms have lower tenured workers (i.e., more new employees) than control firms. Following Caggese et al. (2019), long-tenured workers are defined as worker with tenure longer than 2 years. The proportion of lower tenured workers peaks during the fraud period. Panel B shows that exiting employees at fraud firms earn more wages than newly hired employees. These two findings are consistent with managers at fraud firms changing the mix of employees away from longer tenured, higher paid employees toward newer, lower paid employees.

Table 6 Panel C examines the wage dynamics for these new employees that join during the fraud period. We use a separate sample of "new employees" for this analysis; we require that she *not* work for the sample firm in the year prior to the fraud period, *Pre(t-1)*, and work for the firm for the first year of the fraud period, *Fraud(t)*. New employees at the control firms are also joining in the same, event-time year. New employees have negative wage effects in the post fraud period

[28] These statistics are from the 2016 version of Graham et al. (2019).

[29] McNichols and Stubben (2008) find overinvestment in capital expenditures but find weaker overinvestment with R&D expenditures. They suggest that R&D reduces profits immediately, making it a less attractive type of investment to improve firm performance while perpetrating fraud. Employee wages are similar to R&D: overinvestment in labor would be expensed presently. Shifting the worker composition toward cheaper, new employees could cause the firm to have the appearance of growth without the income statement expense. We confirm the findings of overinvestment in capital expenditures during the fraud in Internet Appendix Figure 5.

(significant for *Post(t+4)*, *Post(t+5)*, and *Post(t+6)*) in the range of -5% and -8% per year. Present value calculations show cumulate wage losses of 30% in this period. Unlike specifications for existing employees, new employees have greater point estimates for the present value at the beginning of the fraud than at the end (-.215 is greater than -0.300); however, these are not statistically different from each other. The sophisticated aspects of these results are worth to mention. Although new employees find an opportunity to work at a firm, the present value estimates suggest that new employees are not fully compensated on the potential displacement risk. Conditional on new employees knowing of the fraud, detection rates would have to be less than 15% to have a positive expected wage outcome, much lower than detection rates estimated from other papers (e.g., 27% or 52% from Dyck et al., 2020). This suggests that new workers lack some information about the prospects of the firm.

After a fraud is revealed, there are three, straightforward reasons which explain why firms will use less labor, i.e., unwind this overbuilding of the workforce. First, conditional on this excess hiring during the fraud period, firms will reduce this (inefficient) hiring when the fraud concludes (Kedia and Philippon, 2009). Second, accounting fraud indicates some governance failure at the firm. Afterward, boards or shareholders could take away decision rights from executives and undertake projects with more caution, causing use of employee labor (and other inputs) to contract (Farber, 2005). Third, Schrand and Zechman (2012) show that excessive optimism (covering up small shocks) tends to precede fraud, which can unravel afterward if the shock worsens. Naturally, a firm's use of labor will decline with a negative shock, especially when the shock causes the firm to fail. Each of these effects would likely cause worker displacements as the firm contracts in the post-fraud period.

Returning to Figure 3, we find negative employee growth in the post fraud period for these fraud firms, again comparing with the control firms and the industry average. After the fraud, employee growth rates are negative; fraud firms' rates average -3%, while control firms average 0% and the industry averages 3%. Like the positive growth rates during the fraud indicating net hiring, negative growth rates indicate that the fraud firms are shedding employees after the fraud period (i.e., the fraud firms have net departures of workers after fraud revelation), while control firms are neither shedding nor hiring and the average firm in the industry is hiring. The post period in the figure corresponds with Table 4 Panel B, where existing employees are displaced from firm, industry, and county at much higher rates relative to existing employees at the control firms.[30]

5.2. Mechanism: Simulation of the Executive's Fraud Decision

Presumably, executives commit fraud considering (expected) benefits and costs (Becker, 1968). Given some executives may argue that fraud would protect their workforce from layoffs or even firm failure, we evaluate the net employee benefits and costs of fraud using reasonable parameter assumptions (John et al., 1992; Couch and Placzek, 2010; Dyck et al., 2020). With this back-of-the-envelope approach, we also estimate the subjective probability of detection where the net cost (and net benefit) to employees is zero and the expected costs for workers of all fraud, both caught and uncaught.

These net costs and subjective probabilities of detection (i.e., likelihood of the fraud going uncaught) calculations are described in Appendix C. Moreover, we ignore all other costs and benefits attributable to fraud not specific to the (potential) fraud firm's workforce. These

[30] In the Internet Appendix Figure 1, we separate out frauds by length of the fraud, i.e., one-year, two-year, and three-year (or more) frauds. One- and two-year frauds have patterns very consistent with the main results and our interpretation. Three-year and longer frauds exhibit long, consistent declines in employee growth and wages during and after the fraud. These long-lasting frauds seem less consistent with cases of overoptimism (e.g., Schrand and Zechman, 2012) and, thus, overbuilding.

calculations require material assumptions that cannot be validated with our data because uncaught fraud (or an unchosen alternative, like layoffs) is hard to measure. However, we borrow measures and estimates from prior literature for, e.g., layoffs at distressed firms (John et al., 1992) and fraud detection rates (Zakolyukina, 2018; Dyck et al., 2020). With a variety of assumptions, only the more optimistic (i.e., lowest chances of being caught and longest lasting benefits to newly hired employees) produce estimates that are potentially beneficial for employees. An example calculation suggests that with avoided layoffs of 5%, continued abnormal growth for 3 years if uncaught, and fraud detection rates of 52% have a cumulative expected cost for workers of 20% of annual wages. Fixing the expected cost at zero indicates that executives expect the probability of (AAER) fraud detection to be 15% or lower, much more favorable than estimates from academic literature (e.g., Dyck et al., 2020).

This simulation of the executive's fraud decision contains another useful calculation nested within: the expected cost to employees of all fraud, both caught and uncaught. As with other papers examining fraud, we only observe detected (or caught) fraud. Importantly, a finding that rank-in-file employees suffer wage losses when firms are discovered to have committed fraud does not necessarily imply that rank-in-file employees suffer, on average, when firms commit fraud because some firms commit fraud and go undetected (or uncaught). In the simulation, the executive is comparing the net cost or net benefit of fraud to an alternative cost, like immediate layoffs. If we drop this alternative cost, the remaining estimate is the cost to employees of all fraud. Across the scenarios presented in Appendix C, the net benefit of all fraud for workers ranges between 2% and -34% (untabulated). The assumptions above (3 years of continued abnormal growth and detection rates of 52%) result in the average employee, between both newly hired and existing, incurring 27% in expected, cumulative wage losses.

5.3. Market Heterogeneity: Thick and Thin Markets

To understand better the source of these wage changes, we descriptively split the result by the character of the market where the employee works. Moretti (2011), in reviewing local labor markets, points out that thick labor markets provide insurance to workers (and firms) against idiosyncratic shocks. He writes, "The presence of a large number of other employers implies a lower probability of not finding another job." This intuitive logic resonates in fraud cases that are particularly harmful to small communities like how the WorldCom's fraud affected Clinton, MS (e.g., Bayot, 2002). We expect the consequences of these frauds in thin labor markets to be particularly devastating for workers who do not have many other employer options.

We separately examine "thick" and "thin" labor markets, i.e., regions with many and few industry-specific employees, respectively. Table 7 shows this sample split in columns 1 and 2. In column 1, we present estimates where the local labor market has many industry-specific employees, i.e., thick labor markets (e.g., Dore and Zarutskie, 2017; Graham et al., 2019).[31] In the post fraud period, the fraud firms in thick labor markets pay less than the control firms, but only two coefficients are statistically significant. The estimated magnitudes ranging between -6% and -9%, through *Post(t+6)*. In column 2, we present estimates where the local labor market has few industry specific employees, i.e., thin labor markets. Employees in these labor markets do very poorly. The negative wage effects in the fraud and post fraud periods are large, e.g., point estimates more negative than -10% for years after *Post(t+4)*. We depict these effects in Figure 4 Panel A. Overall, the wage declines are *much* stronger in thin labor markets, indicating that the effect seems to come from displacement into crowded labor markets and frictions to effective job-searches (e.g.,

[31] In untabulated tests, we also measure thick labor markets based on the size of industry-specific labor demand (i.e., the number of employers). Our results are qualitatively the same with this alternative sample split.

Moretti, 2011). This result has an important implication for SEC enforcement. Kedia and Rajgopal (2011) find that SEC enforcement actions concentrate on firms in big cities. However, some negative consequences are more severe for fraudulent firms elsewhere, i.e., where there are thin labor markets.

5.4. Firm Heterogeneity: Bankruptcy and Non-Bankruptcy

Another source of variation that is relevant for understanding the consequences for employees is the seriousness of the fraud or seriousness of the shock that the fraud is hiding. Many big frauds can be associated with firm failure, e.g., Enron in late 2001 and early 2002 (SEC, 2004). In addition, Graham et al. (2019) examine the wage effects of bankruptcy (independent of fraudulent reporting) and find negative consequences for employees in the post-bankruptcy period. To put these analyses in context, some descriptive statistics are useful. 5% of firms go through a bankruptcy (e.g., Hillegeist et al., 2004), and we calculate that 10% of all AAER fraud firms go through a bankruptcy. So, while only twice as likely to go bankrupt compared with non-fraud firms, perhaps this is the relevant subsample driving employee wage and turnover consequences. We want to both (i) see if the consequences vary with seriousness of the fraudulent misreporting and (ii) determine whether serious firm distress or firm failure can fully explain our results.

To provide evidence on this variation, we examine bankruptcy and non-bankruptcy fraud firms using the UCLA-LoPucki Bankruptcy Research Database, which includes cases filed under Chapter 7 and Chapter 11. For this subsample analysis, we retain the matched-control firm for bankrupt and non-bankrupt fraud firms. Table 7 shows this sample split in columns 3 and 4. In column 3, we present estimates where the fraud firm declares bankruptcy within three years after the fraud period. Employees of bankrupt fraud firms have declines in wages during *Fraud(t)*. Subsequently, there are large drops in wages. The magnitudes in the post fraud period range

between -15% and -22%, recovering in the later years. Wage drops for employees of bankrupt fraud firms is severe. In column 4, we present estimates where the fraud firm does not declare bankruptcy within three years after the fraud period. Again, there are negative wage effects in the post fraud period, though are less severe than the bankruptcy subsample, fluctuating between -6% and -13% (and significant at conventional levels in years *Post(t+4)*, *Post(t+5)*, and *Post(t+6)*). We also depict these effects in Figure 4 Panel B. Note that the observation count for this non-bankruptcy subsample is the majority of our full sample (i.e., 93.7%). While devastating, bankruptcies do not drive the overall wage decline from fraudulent financial reporting in our main analysis, even employees at fraud firms without bankruptcy suffer negative consequences.[32] These findings are consistent with fraud devastating employee' labor market outcomes via another channel, overbuilding.

5.5. Employee Heterogeneity: Movements and Pre-Fraud Wages

A final source of variation that can help inform why employees suffer these negative wages around fraud comes from the employees themselves. First, we show descriptive splits based on worker movements both at fraud and matched control firms. We examine the subsample of employees who leave before three years in the post-fraud period ("leaver"); this subsample includes leavers from both fraud and matched control firms. These results are in column 1 of Table 8. Leavers of fraud firms experience a steady decline in wages during the fraud period. Second, we examine the subsample of employees who stay through at least three years in the post-fraud

[32] We can compare these magnitudes to Graham et al. (2019) who examine the wage effects of bankruptcy (independent of fraudulent reporting). They find that cumulative wage losses are about 84% for bankruptcy. As a rough comparison, the wage consequences for employees are greater when the executives commit fraud and file for bankruptcy at 96%. More to the point, analyses in the Internet Appendix Table 3 show that relative to non-fraud-bankruptcy firms, fraud-bankruptcy firms' employees' cumulative wage losses are larger by 39%, which is comparable to the main results in Table 3.

period ("stayer"), in Table 8 column 2. While all coefficients in the fraud and post fraud period are negative, these coefficients are not significant at conventional levels, indicating that stayers at fraud firms may have similar wage trajectories as stayers at control firms. These stayers provide evidence that employee separations are an important factor in wage consequences for fraudulent financial reporting.

In Figure 5 Panel A, we perform an alternative split to compare workers within the fraud firm conditional on displacement timing. We separate employees who (i) stay through at least three years in the post-fraud period ("stayer"), (ii) leave in the first year of the fraud period ("early leaver"), and (iii) leave after the first year of the fraud period but before three years in the post-fraud period ("late leaver"). The figure shows an interesting dynamic for fraud-firm employees who are early versus late leavers. Early leavers experience negative wage effects during the fraud period (i.e., when they leave the fraud firm) but afterward experience a recovery of wages with cumulative wage losses equal to 34% (discounted to *Fraud(t)*).[33] Late leavers, on the other hand, have negative wage effects in both the fraud and post-fraud periods with cumulative wage losses of 117%, which is consistent with two features of accounting fraud. First, conditional on leaving, workers would likely prefer to be early leavers unless they were unaware of the fraud or believe detection rates are below 22%. Second, revelation causes disruption to local labor markets, i.e., the reversal of overbuilding has severe consequences. Late leavers' wage declines again show, like results for thin labor markets, that the effect likely comes from displacement into crowded labor

[33] We discount to *Fraud(t)* to simulate the timing of an employee deciding when to leave the firm. For completeness, stayers have cumulative wage gains equal to 45% compared to the average worker at the control firm (who may stay or may leave). These analyses show that workers who stay at the fraud firm do better than workers who leave. Overall, we conclude that worker displacement is an important driver of negative consequences for workers from fraudulent financial reporting.

markets, negatively affecting the leavers' job searches (e.g., Moretti, 2011). These negative wage effects for late leavers are persistent through the end of the series.

Finally, we use pre-fraud variation in pay to provide some evidence on whether these are wage effects for the highest paid employees (e.g., potentially executives or culpable accountants being penalized for the fraud) or if lower paid employees also suffer wage drops around fraudulent financial reporting. This analysis is rarely found in prior papers measuring wage losses (e.g., Jacobson et al., 1993; Hummels et al., 2014), but it is important in our context because of the culpability issue. We split out the top 10% from other workers following prior literature (e.g., Autor et al., 2008).

For columns 3 and 4 in Table 8, we present analyses that condition on the pre-fraud period wage level across firms, splitting the sample into workers who are in the top 10% of the wage distribution ("top 10%") and the bottom 90% of the distribution ("bottom 90%"). In column 3, employees in the top 10% do not suffer significant negative consequences during or after the fraud period. Bottom 90% employees, however, experience significant, negative wage effects in the fraud and post-fraud periods, as severe as -7% during the fraud and between -6% and -13% after the fraud. Workers in the bottom 90% of the wage distribution have worse wage consequences from fraudulent financial reporting despite the lower likelihood that they are involved with the fraud.[34] Although fraudulent financial reporting decisions are made at the management level, the results indicate that the costs are borne mainly by non-management workers. Low wage workers suffer more even though they are not likely the executives who commit fraud. In addition, this analysis has some connection to the thick and thin labor markets analysis, which is about the

[34] We define the top 10% of the wage distribution within industry. Different methods for calculating the wage distribution, e.g., measuring the wage distribution across industries, do not change the results qualitatively.

market into which the workers are being displaced. The bottom 90% can be the employee side of the same story. These workers could face more job search frictions in labor markets with less portable skills.[35] We depict these results in Figure 5 Panel B.

6. Conclusion

This paper provides evidence on the consequences for employees from fraudulent financial reporting. We use employer-employee matched data from the U.S. Census Bureau combined with SEC enforcement actions against firms with serious misreporting ("fraud") to examine wages and employee turnover. Compared to the employees at non-fraud control firms, we find that employees at fraud firms have lower wages after periods of fraudulent financial reporting even though fraud firms have higher employment growth during the fraud. During the fraud, executives appear to overbuild and change employee composition. Also, we find that employees at fraud firms are more likely, compared to a matched sample, to leave the firm, industry, and (even) county of employment after the fraud is revealed. Overall, fraud firms unwind this overbuilding and have negative employee growth, likely driving the negative wage consequences.

We discuss and show evidence consistent with mechanisms for these wage effects. The negative change in wages combined with employee displacement and negative employment growth at fraud firms indicates workers suffer negative labor market outcomes, for instance losses of firm-specific investments, job search inefficiency, and/or entering crowded labor markets. Wage losses are worse in thin labor markets and for fraud firms that ultimately declare bankruptcy.

[35] Hass et al. (2019) find that worker mobility can discipline executives' myopic decisions. Applying this idea to our setting, perhaps executives anticipate employee consequences in their fraud decisions. Thus, the firms with employees who will have the worst outcomes are more likely to commit fraud to avoid layoffs. On the other hand, we find workers who have high mobility, e.g., those in thick labor markets or with high wages, have the lowest costs, so we may see executives commit fraud more often when they have highly mobile workers who would not be damaged by the fraud.

However, employees of non-bankrupt fraud firms also suffer wage declines, so the effects are not isolated to failed firms. Finally, we examine workers in the bottom 90% (top 10%) of the pre-fraud wage distribution and perhaps surprisingly (do not) find negative wage effects during and after fraudulent financial reporting.

This paper is among the first to measure wage and turnover effects for employees at fraud firms. We note some caveats and potential avenues for future research. First, we show evidence that is consistent with certain real actions taken by executives that relate to fraud, like overbuilding during the fraud; however, isolating the specific channel is a challenge and another (unexplored) channel could exist. For instance, the job search frictions after the fraud and disruption to labor markets are both related to the severity of economic shocks to the firm. Second, matched difference-in-differences designs do not necessarily show causation (Roberts and Whited, 2013). We find the labor market consequences that happen concurrently, with little evidence for pre-period trends, so we are confident these effects are associated with the fraud but not necessarily caused by it. Third, the (unexplored) heterogeneous consequences of fraudulent financial reporting for employees might deserve more attention. Furthermore, we do not study the effects spreading beyond the affected employees (e.g., their families and neighborhoods). Finally, SEC enforcement priorities could focus on more severe fraud rather than neutrally target cases of serious misreporting; other types of misreporting could be consequential as well but not in our sample.

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Appendix A: Variable Definitions

Variable	Definition	Data Source
Dependent Variables		
Annual Real Wages	Annual earnings from a primary employer divided by the Consumer Price Index (2010)	LEHD
Fraud Firm Indicator	Companies that are identified as accounting-fraud firms by the AAER from 1970 through 2014	CFRM, AAERs
Independent Variables		
Fraud Indicator	Workers who are at fraud firms as either an *Existing Employee* or a *New Employee*	LEHD
Pre(t-p)	1 if year τ falls p(=1,2,3,4) year(s) before a fraud firm engaged in accounting fraud; 0 otherwise	CFRM, AAERs
Fraud(t+p)	1 if year τ falls p(=0,1,2) year(s) after the first year of accounting fraud and if a fraud firm engaged in accounting fraud in year τ; 0 otherwise. For long-lasting frauds, we normalize this period to a maximum of three years by indicating additional fraud years as *Fraud(t+2)*	CFRM, AAERs
Post(t+p)	1 if year τ falls p(=3,4,5,6,7,8) year(s) after the first year of accounting fraud, normalized so that p=3 is the year after an accounting fraud is revealed; 0 otherwise	CFRM, AAERs
Sample Splits		
Existing Employee	Worker at a fraud or control firm for the last two years before a fraud firm engaged in accounting fraud, Pre_{t-2} and Pre_{t-1}	LEHD
New Employee	Worker newly hired in the first year of a fraud period, $Fraud_t$, by a fraud or control firm	LEHD
Long-tenured Employee	Worker who has worked for a company more than 2 years	LEHD
Thin / Thick Labor Markets	Thin local labor markets have fewer industry-specific employees than the median of the number of industry-specific employees in local labor markets	LEHD
Bankruptcy Fraud Firms	Fraud firms that declare bankruptcy within three years after frauds are revealed	BRD
Stayer / Leaver	Stayer if an employee continues to work for the fraud or control firm three years after the accounting fraud is revealed, $Post_{t+6}$ and/or later; leaver otherwise	LEHD
Early / Late Leaver	Early leaver if an employee left the fraud or control firm in the first year of accounting fraud, $Fraud_t$; late leaver if the fraud or control firm in any other year of accounting fraud or within two years after accounting fraud is revealed, $Fraud_{t+1}$ through $Post_{t+5}$	LEHD
Top 10%	Workers earn real wages more than or equal to the 10 percentile real wage in the wage distribution by industry	LEHD
Bottom 90%	Workers earn real wages less than the 10 percentile real wage in the wage distribution by industry	LEHD

Appendix A: Variable Definitions (continued)

Variable	Definition	Data Source
Firm Controls		
Size	Natural log of total assets (data6)	Compustat
Return on Assets	Operating income after depreciation (data178) divided by total assets (data6)	Compustat
Leverage	The ratio of total debt (data9+data34) to market value of assets, which is calculated by multiplying the number of shares outstanding (data25) by the stock price (data199) and by adding total debt (data9+data34) to it	Compustat
Tobin's Q	Market value of assets divided by book value of assets (data6), where market value of assets is calculated by (data25*data199+data9+data34)	Compustat
Employee Growth	Natural log of this year's employment minus natural log of last year's employment	LBD
Avg. Annual Real Wages	Total wage bill divided by employment	LBD
Employee Controls		
Age	Age of an employee in an event year of accounting fraud	LEHD
Education	Four levels of education are transformed into numerical values by using the highest number of years in each category: less than high school (1-11), high school or equivalent, no college (12), some college or associate degree (13-15), and bachelor's degree or advanced degree (16-19)	LEHD
Experience	Age of a worker in year t minus education minus 6	LEHD
Female	1 if a person is female; 0 otherwise	LEHD

Appendix B: Probit Model

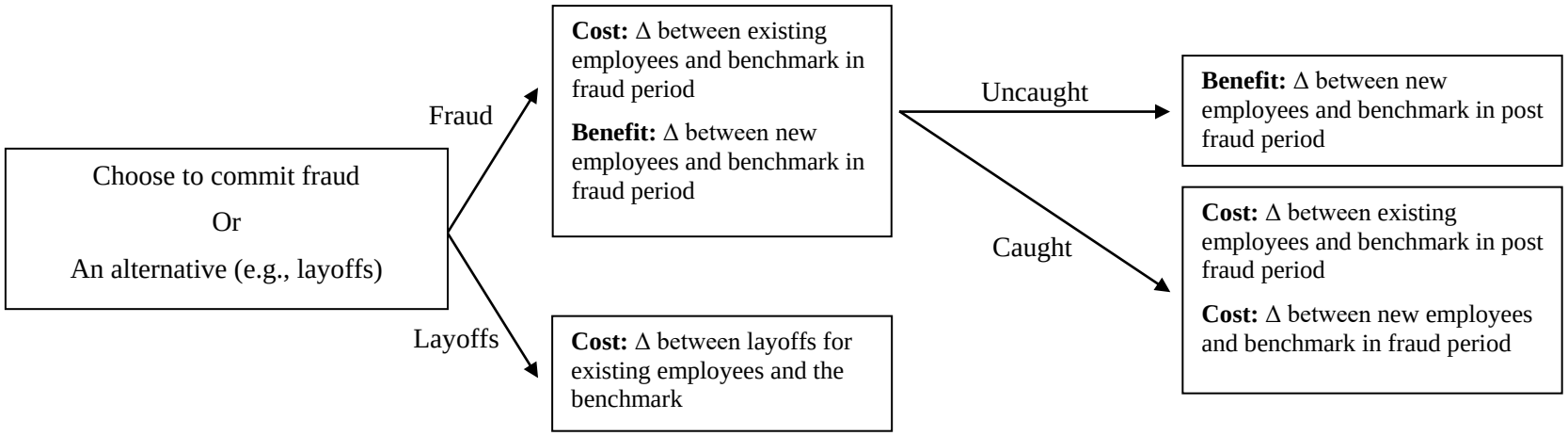
This table shows the results of a probit model estimating a propensity score to engage in accounting fraud. Accounting-fraud firms are identified by the AAER. Fraud firms are included in sample firms in the year prior to accounting fraud, *Pre(t-1)*. Non-fraud firms are included in sample firms if they operate businesses in the same industry as one of fraud firms in the year prior to accounting fraud. The sample period is from 1989 to 2008. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1)
Dependent Variable: *Fraud-Firm Indicator*	
Size	0.061***
	(0.016)
Sales Growth	0.375***
	(0.085)
Return on Assets	0.455**
	(0.199)
Leverage	-0.029
	(0.113)
Tobin's Q	0.018**
	(0.008)
Employee Growth	0.096
	(0.076)
Ln*(Avg. Annual Real Wages)*	-0.064
	(0.051)
Observations	30,000
Chi-squared	55.49
Pseudo R-squared	0.028

Appendix C: Worker-Focused Rationale for Committing Fraud

Panel A: Diagram of Fraud Decision Tree and Relevant Benefits and Costs

This diagram gives the manager's decision tree to commit fraud weighing the ex ante present value (PV) for workers. The benefit and cost inputs to the PV calculation are derived only from benefits or costs to workers and ignore any other benefits or costs to the manager, shareholders, other firm counterparties, etc. We approximate these benefits or costs relative to a benchmark, either the matched firm from our empirical analyses in the body of the paper or the industry average. We identify these as benefits or costs when the outcome appears to be positive or negative, respectively, relative to the benchmark for existing employees (those at the firm when the fraud starts) or for new employees (those who join the firm after the fraud starts). There are three exogenous parameters in the PV calculation, (i) the cost of the alternative option to fraud, (ii) the probability of getting caught committing fraud, and (iii) the length of time needed for the fraud firm to revert to the benchmark. We assume that the alternative option to fraud is laying off workers, which we calculate as the proportion of existing employees who depart from the firm multiplied by wage losses from the layoffs. Also, we assume that a fraud firm's reversion to the benchmark only has benefits for new employees (i.e., continued time with higher than benchmark employee growth) and has no continuing costs for existing employees (i.e., no continued departures nor wage losses for existing employees relative to the benchmark).

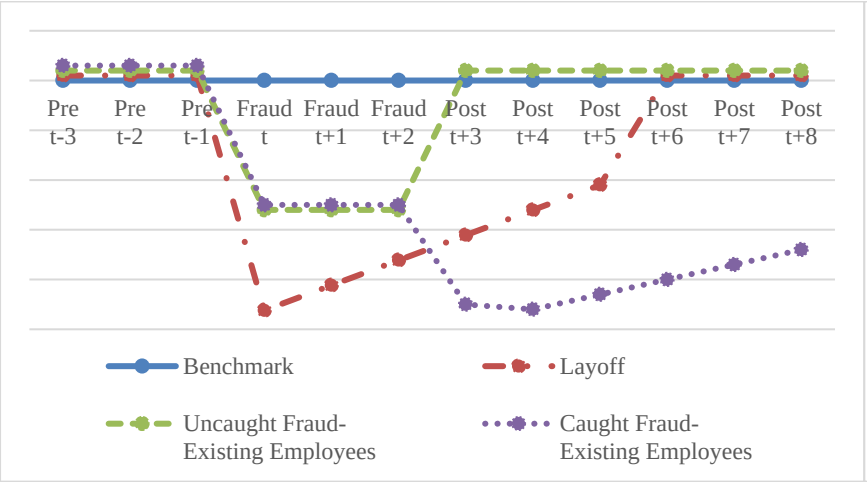


Appendix C: Worker-Focused Rationale for Committing Fraud—continued

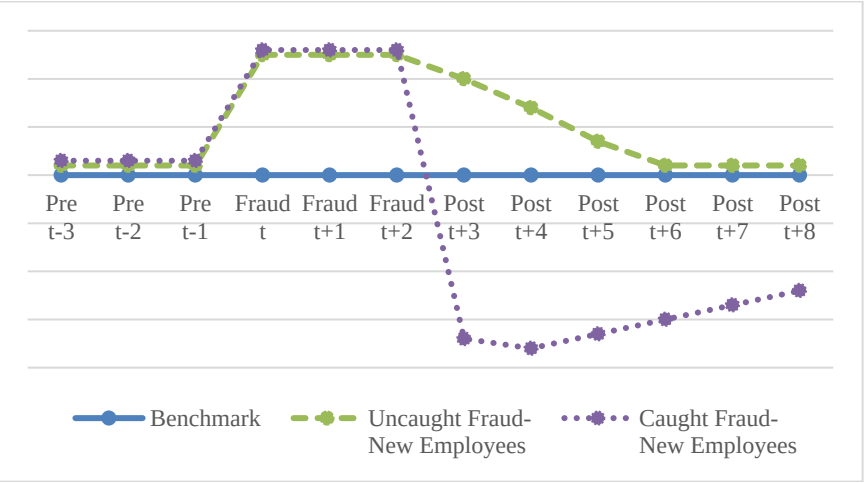
Panel B: Figures with Hypothetical Benefits and Costs in Event Time

These figures provide representations of the benefits and costs described in Panel A. For existing employees, in Figure B.1, the alternative is the cost imposed by layoffs relative to the benchmark, i.e., the difference between the dot-dash line and the solid line. Existing employees incur the cost of the fraud, prior to being caught or uncaught, is the difference between the dot line (or dash line) and the solid line during the fraud period. If the fraud is caught, existing employees incur additional costs, i.e., the difference between the dot line and the solid line in the post fraud period. For new employees, in Figure B.2, benefits from the fraud are received, prior to being caught or uncaught, i.e., the difference between the dot line (or dash line) and the solid line during the fraud period. If the fraud is uncaught (caught), new employees enjoy additional benefits (incur costs), i.e., the difference between the dash line (dot line) and the solid line in the post fraud period.

B.1. Existing Employees



B.2. New Employees



Appendix C: Worker-Focused Rationale for Committing Fraud—continued

Panel C: Tabulation of Ex Ante Present Values for Workers

This table reports an example ex ante present value (PV) for workers to commit fraud as a percentage of annual wages. Benefit and cost inputs to the PV calculation are derived from our analyses, prior literature, and flexible assumptions. The discount rate is equal to PV calculations elsewhere in the paper, i.e., 4.47%. For this example, the calculation uses an alternative scenario where 5% of workers depart as opposed to committing fraud, i.e., an avoided cost with fraud is not laying off 5% of the existing employees (John et al., 1992). Lost wages for departing workers are calculated over a 6-year period using initial and final year wage loss estimates from Couch and Placzek (2010) with linear interpolation. The chance of getting caught is the estimate from Dyck et al. (2020) for caught AAERs. This example assumes that the manager expects a 3-year horizon when reverting from excess hiring during the fraud period to the benchmark in the post fraud period. Hired new employees avoid lost wages from unemployment, equal to the mean unemployment rate during our sample period of 6.14%. The benchmark case for these calculations are the matched control firms. We denote an annuity function "PV" with arguments of the discount rate, the number of periods, and the percentage of annual wages, respectively.

(1)	(2) Source or Assumption	(3) Calculation	(4) Subtotal
Fraud			
Cost: existing employees	T3:PA:C(1)	-52.4% + 47.5% / (1+4.47%)^3	-10.7% (A)
Benefit: new employees	T8: C(5); F3: PB- 19.5% [Cumulative New Employees]	(30.0% / (1+4.47%)^3 - 21.5%) * 19.5% + 6.14% * PV(4.47%, 3, 100%) * 19.5%	4.2% (B)
Uncaught Fraud			
Benefit: new employees	F3: PB- 19.5% [Cumulative New Employees]	6.14% * PV(4.47%, 6, 100%) * 19.5% / (1+4.47%)^3	5.4% (C)
Benefit: additional post fraud new employees	Assume 3-year reversion to steady state growth; F3: PB- 9.1% [Avg Emp. Growth during Fraud] 0.2% [Avg Control Emp. Growth after Fraud]	(9.1% - 0.2%) * (3/4) / (1+4.47%)^3 * 6.14% * PV(4.47%, 6, 100%) + (9.1% - 0.2%) * (2/4) / (1+4.47%)^4 * 6.14% * PV(4.47%, 5, 100%)+ (9.1% - 0.2%) * (1/4) / (1+4.47%)^5 * 6.14% * PV(4.47%, 4, 100%)	3.3% (D)
Caught Fraud			
Cost: existing employees	T3:PA:C(1)	-47.5% / (1+4.47%)^3	-41.7% (E)
Cost: new employees	T8: C(5); F3: PB- 19.5% [Cumulative New Employees]	-30.0% / (1+4.47%)^3 * 19.5%	-5.1% (F)
Alternative			
(Avoided) Cost: existing employees	Assume 5% layoffs (John et al.,1992) and 33% wage declines initially and 15% after six years (Couch and Placzek, 2010)	5% *[(33% * 5/5 + 15% * 0/5) / (1+4.47%)^1 + (33% * 4/5 + 15% * 1/5) / (1+4.47%)^2 + … + (33% * 0/5 + 15%* 5/5) / (1+4.47%)^6]	6.3% (G)
Total NPV for Fraud	Assume 52% chance of getting caught (Dyck et al., 2020)	(A) + (B) + (G) + 52% * ((E) + (F)) + (1 - 52%) * ((C) + (D))	-20.3%

Appendix C: Worker-Focused Rationale for Committing Fraud—continued

Panel D: Tabulation of Ex Ante Present Values for Workers

This table reports estimates of the ex ante present value (PV) for workers to commit fraud as a percentage of annual wages. Benefit and cost inputs to the PV calculation are derived from our analyses, prior literature, and flexible assumptions. The discount rate is equal to PV calculations elsewhere in the paper, i.e., 4.47%. Also presented are subjective probabilities an uncaught fraud where the PV equals zero; if the breakeven chance is calculated as negative, we indicate that the probabilities are not meaningful ("nm"). We use an alternative scenario where 5% of workers depart as opposed to committing fraud, i.e., an avoided cost with fraud is not laying off 5% of the existing employees (John et al., 1992). Lost wages for departing workers are calculated over a 6-year period using initial and final year estimates from Couch and Placzek (2010) with linear interpolation. Column (1) indicates the chance of getting caught (latent fraud detection rates) using various estimates from Zakolyukina (2018) and Dyck et al. (2020). Columns (2)-(4) and (5)-(7) use match control firms and an industry average, respectively, as the benchmark (counterfactual) for estimating normal employment growth. Columns (2) & (5), (3) & (6), and (4) & (7) calculate benefits for new employees with assumptions of 1-, 3-, and 5-year horizons, respectively, when reverting from excess hiring during the fraud period (using the gross, average employment growth of 9.1%) to the benchmark in the post fraud period.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Benchmark Firm:		**Matched Control Firms**			**Industry Average**		
Time to revert to steady state if fraud is not caught:		1 year	3 years	5 years	1 year	3 years	5 years
Estimations of latent detection rates of accounting misrepresentation in—	Chance caught:	**Ex ante PV of fraud for average worker as a percentage of annual wages**					
Zakolyukina (2018) caught misstatements in any one year	3%	4.9%	6.9%	8.3%	-3.7%	-2.3%	-1.3%
Zakolyukina (2018) caught misstatements over 5 years	16%	-2.1%	-0.4%	0.8%	-10.1%	-8.9%	-8.0%
Dyck et al. (2020) caught fraud by auditor	29%	-9.0%	-7.6%	-6.5%	-16.3%	-15.3%	-14.6%
Dyck et al. (2020) caught AAER fraud by SEC	52%	-21.3%	-20.3%	-19.6%	-27.4%	-26.8%	-26.3%
		Subjective probabilities of uncaught fraud with breakeven PV (i.e., ex ante PV = 0%)					
		12.1%	15.4%	17.5%	nm	nm	0.4%

Appendix D: Illustration of Present Value Calculation

This table reports the approach to the present value calculations, using coefficient estimates from Table 3 Column 1 in this Appendix Column (1). Coefficient estimates are converted to percentage wage loss estimates in Column (2). We calculate two present values. Columns (3) and (4) contain present values calculated for the end of the pre fraud and pre revelation period, respectively. The calculations use inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%).

		(1) Coefficients from Table 3 Column 1	(2) Conversion to Percentages	(3) Pre Fraud (t-1) Present Value	(4) Pre Revelation (t+2) Present Value
	A	B	C = Exp(B) – 1	C / (1 + 4.47%)^A	C / (1 + 4.47%)^(A – 3)
Fraud(t)	1	-0.024	-2.3%	-2.2%	
Fraud(t+1)	2	-0.032	-3.2%	-2.9%	
Fraud(t+2)	3	-0.066	-6.4%	-5.6%	
Post(t+3)	4	-0.058	-5.6%	-4.7%	-5.4%
Post(t+4)	5	-0.091	-8.7%	-7.0%	-8.0%
Post(t+5)	6	-0.117	-11.0%	-8.5%	-9.6%
Post(t+6)	7	-0.130	-12.2%	-9.0%	-10.2%
Post(t+7)	8	-0.098	-9.3%	-6.6%	-7.5%
Post(t+8)	9	-0.092	-8.8%	-5.9%	-6.7%
Total				-52.4%	-47.5%

Table 1. Fraud Firms

Panel A. Comparison of Sample Fraud and Matched Control Firms

This table compares fraud firms' to control firms' characteristics in the year prior to accounting fraud, *Pre(t-1)*. Accounting-fraud firms are identified by the AAER. Control firms are matched to fraud firms based on a propensity score estimated in Appendix Table B. The sample period is from 1989 to 2008. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1) Fraud Firms	(2) Control Firms	(3) T Tests of Differences (Fraud minus Control)	
			Difference	Standard Error
Size	6.482	6.310	0.172	(0.226)
Assets ($M)	5,550	3,359	2,191*	(1,266)
Sales Growth	0.228	0.237	-0.009	(0.036)
Return on Assets	0.080	0.087	-0.007	(0.012)
Leverage	0.257	0.233	0.024	(0.027)
Tobin's Q	2.309	2.259	0.050	(0.220)
Employee growth	0.140	0.151	-0.011	(0.045)
Avg. Annual Real Wages ($Th)	50.34	49.73	0.610	(3.905)
Employees (Headcount in U.S.)	10,440	8,961	1,479	(1,908)
Observations	150	150		

Table 1. Fraud Firms

Panel B. Comparative Descriptive Statistics on Sample and All Fraud Firms

This table compares statistics on samples of fraud firms. Column (1) indicates descriptive statistics of sample fraud companies, and column (2) indicates descriptive statistics of all fraud firms. Fraud firms are identified by the AAER. All fraud companies are required to have relevant Compustat data. They engaged in accounting fraud from 1970 to 2014. Sample fraud companies are required to have relevant Compustat, LBD, and LEHD data. They engaged in accounting fraud from 1989 to 2008. Appendix Table A defines variables. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1) Sample Fraud Firms	(2) All Fraud Firms
Size	6.482	5.423
Assets ($M)	5,550	4,102
Return on Assets	0.080	.0157
Leverage	0.257	.2552
Tobin's Q	2.309	2.838
Observations	150	500

Table 2. Descriptive Statistics for Employees of Fraud and Control Firms

This table shows differences for averages of employees at fraud and control firms. Accounting-fraud firms in the sample commit financial misrepresentation from 1989 to 2008 according to the AAER. Fraud firms are matched with control firms using a propensity score estimated in Appendix Table B. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1) Fraud Firms	(2) Control Firms	(3) T-Test of Differences (Fraud minus Control)	
			Difference	Standard Error
Education	14.20	13.77	0.43	(0.279)
Age	37.90	36.68	1.22	(0.880)
Experience	17.70	16.91	0.79	(0.695)
Annual Real Wages	71,210	54,220	16,990**	(8,310)
Female	0.449	0.439	0.01	(0.060)
Observations	361,800	404,400		

Table 3. Dynamics and Present Values of Earnings for Fraud Firm Employees

This table reports estimates from OLS regression analyses estimating equation (2), presenting estimates for wage effects at fraud firms in the by-event-time years. Standard errors are in parentheses and calculated with clustering by pre-fraud employer (i.e., fraud firm or matched control firm). At the bottom, we give estimates for present value calculations of wage effects at the end of the indicated event-time period from all subsequent by-event-time year estimates using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from the by-event-time year estimates. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

Dependent Variable = Ln(*Annual Real Wages*)	(1) Year Effects	(2) Year- Industry Effects	(3) Year- Industry-County Effects
Pre(t-3) × Fraud Ind.	-0.012	-0.029*	0.001
	(0.025)	(0.016)	(0.013)
Pre(t-2) × Fraud Ind.	-0.012	-0.042	0.020
	(0.029)	(0.025)	(0.023)
Pre(t-1) × Fraud Ind.	-0.021	-0.063**	-0.007
	(0.032)	(0.030)	(0.026)
Fraud(t) × Fraud Ind.	-0.024	-0.050	-0.007
	(0.026)	(0.031)	(0.021)
Fraud(t+1) × Fraud Ind.	-0.032	-0.057	0.006
	(0.039)	(0.042)	(0.023)
Fraud(t+2) × Fraud Ind.	-0.066	-0.080**	-0.025
	(0.045)	(0.038)	(0.025)
Post(t+3) × Fraud Ind.	-0.058	-0.081**	-0.025
	(0.041)	(0.037)	(0.025)
Post(t+4) × Fraud Ind.	-0.091*	-0.110***	-0.055*
	(0.047)	(0.036)	(0.031)
Post(t+5) × Fraud Ind.	-0.117**	-0.143***	-0.094***
	(0.050)	(0.038)	(0.034)
Post(t+6) × Fraud Ind.	-0.130**	-0.122***	-0.074**
	(0.055)	(0.041)	(0.035)
Post(t+7) × Fraud Ind.	-0.098*	-0.119***	-0.062*
	(0.058)	(0.043)	(0.037)
Post(t+8) × Fraud Ind.	-0.092	-0.140**	-0.107***
	(0.094)	(0.056)	(0.039)
Controls	Yes	Yes	Yes
Fixed Effects	Event-time, Year, Worker	Event-time, Year × Industry, Worker	Event-time, Year × Industry × County, Worker
Observations	8,720,000	8,720,000	8,720,000
R-squared	0.606	0.627	0.656
Pre Fraud (t-1) Present Value	-0.524*	-0.669***	-0.318*
	(-1.82)	(-2.80)	(-1.73)
Pre Revelation (t+2) Present Value	-0.475*	-0.574***	-0.338**
	(-1.95)	(-3.22)	(-2.22)

Table 4. Dynamics of Displacement for Employees

This table shows averages and differences of employee retention at fraud and matched control firms. Panel A gives statistics for the fraud period. Panel B gives statistics for the post fraud period. These indicator variables equal one if an employee stays at the firm, in the industry, or in the county and equal zero otherwise. Specifically, if we observe the employee with their *next* job at the same firm, or in the same industry, or in the same county as the firm where the employee is employed during periods Pre_{t-1} and Pre_{t-2}, then the indicator variable equals one. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

Panel A: Fraud Period

	(1) Fraud Firms	(2) Control Firms	(3) T-Test of Differences (Fraud minus Control)	
			Difference	Standard Error
Fraud(t)				
% Stay at Firm	82.3%	87.3%	-5.0%**	(0.025)
% Stay in Industry	81.5%	88.2%	-6.7%*	(0.035)
% Stay in County	84.8%	87.3%	-2.5%	(0.027)
Fraud(t+1)				
% Stay at Firm	66.2%	68.7%	-2.5%	(0.039)
% Stay in Industry	74.6%	72.3%	2.3%	(0.034)
% Stay in County	72.5%	75.4%	-2.9%	(0.038)
Fraud(t+2)				
% Stay at Firm	47.8%	53.5%	-5.7%	(0.055)
% Stay in Industry	63.0%	57.1%	5.9%	(0.037)
% Stay in County	63.2%	64.2%	-1.0%	(0.037)
Observations	361,800	404,400		

Table 4. Dynamics of Displacement for Employees—continued

Panel B: Post Fraud Period

	(1) Fraud Firms	(2) Control Firms	(3) T-Test of Differences (Fraud minus Control)	
			Difference	Standard Error
Post(t+3)				
% Stay at Firm	49.0%	67.6%	-18.6%**	(0.074)
% Stay in Industry	56.6%	70.6%	-14.0%**	(0.062)
%Stay in County	62.4%	73.5%	-11.1%**	(0.056)
Post(t+4)				
% Stay at Firm	40.8%	59.5%	-18.7%**	(0.073)
% Stay in Industry	50.3%	64.3%	-14.0%**	(0.060)
%Stay in County	56.2%	68.5%	-12.3%**	(0.054)
Post(t+5)				
% Stay at Firm	34.4%	55.2%	-20.8%***	(0.071)
% Stay in Industry	42.7%	59.6%	-16.9%***	(0.063)
%Stay in County	50.8%	64.8%	-14.0%**	(0.056)
Post(t+6)				
% Stay at Firm	31.6%	50.3%	-18.7%**	(0.079)
% Stay in Industry	41.6%	55.8%	-14.2%**	(0.067)
%Stay in County	47.0%	61.5%	-14.5%**	(0.057)
Post(t+7)				
% Stay at Firm	28.1%	44.5%	-16.4%**	(0.076)
% Stay in Industry	38.4%	51.5%	-13.1%**	(0.060)
%Stay in County	44.7%	57.8%	-13.1%**	(0.051)
Post(t+8)				
% Stay at Firm	24.5%	42.0%	-17.5%**	(0.082)
% Stay in Industry	33.1%	49.1%	-16.0%**	(0.064)
% Stay in County	40.3%	56.2%	-15.9%***	(0.058)
Observations	361,800	404,400		

Table 5. Sample Attrition and Different Matching Approaches

This table reports estimates from OLS regression analyses estimating equation (2). Panels A, B, and C use various methods to control for sample attrition. Panel A replaces missing values with the sample's 10th percentile of wages. Panel B replaces missing values with the worker's previous wage (i.e., a lagged wage from the most recent non-missing observation in that worker's series). Panel C uses a comprehensive panel of worker-years; this data indicates if a worker receives wages in any state in the U.S. We replace missing values with (i) 10th percentile of wages if the comprehensive panel indicates that the worker does not receive wages, (ii) the worker's previous wage if the comprehensive panel indicates that the worker does receive wages, and (iii) the sample's 10th percentile of wages otherwise. For Panel D, each column uses separate samples of control employees from matches described in the column header. Standard errors are in parentheses and calculated with clustering by fraud-period employer (i.e., fraud firm or matched control firm). All panels present estimates for present value calculations of wage effects at the end of the indicated event-time period using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from estimates for wage effects at fraud firms in the by-event-time years. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. For PVs, significance at the 10% confidence level for a one-tailed test is indicated by ^. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1)	(2)	(3)
Controls	Yes	Yes	Yes
Fixed Effects	Event-time, Year, Worker	Event-time, Year × Industry, Worker	Event-time, Year × Industry × County, Worker
Observations	8,720,000	8,720,000	8,720,000
Panel A: Missing Worker-Years Replaced with 10th Percentile Wages			
Pre Fraud (t-1) Present Value	-0.500**	-0.515**	-0.184
	(-2.03)	(-2.35)	(-0.919)
Pre Revelation (t+2) Present Value	-0.523**	-0.507***	-0.276*
	(-2.44)	(-3.17)	(-1.72)
Panel B: Missing Worker-Years Replaced with Lagged Worker-specific Wage			
Pre Fraud (t-1) Present Value	-0.522*	-0.692**	-0.226
	(-1.65)	(-2.50)	(-1.19)
Pre Revelation (t+2) Present Value	-0.500*	-0.616***	-0.283*
	(-1.81)	(-2.87)	(-1.78)
Panel C: Nationwide Worker-Year Panel to Replace Missing Worker-Years with Lagged Worker-specific Wage			
Pre Fraud (t-1) Present Value	-0.532*	-0.672***	-0.320*
	(-1.89)	(-2.76)	(-1.74)
Pre Revelation (t+2) Present Value	-0.485**	-0.582***	-0.344**
	(-2.04)	(-3.21)	(-2.26)

Table 5. Sample Attrition and Different Matching Approaches—continued

Panel D: Different Matching Approaches

Dependent Variable = Ln(*Annual Real Wages*)	(1) Fraud Period, Unmanaged Sales Growth	(2) Within-Firm Employee Characteristics	(3) Employee Match within Industry	(4) Revelation Period, *Post (t+3)*, Match
Pre(t-3) × Fraud Ind.	-0.000	-0.026*	0.002	0.023
	(0.024)	(0.015)	(0.017)	(0.018)
Pre(t-2) × Fraud Ind.	-0.004	-0.007	-0.014	-0.005
	(0.033)	(0.024)	(0.014)	(0.025)
Pre(t-1) × Fraud Ind.	-0.031	-0.022	-0.036*	-0.033
	(0.039)	(0.029)	(0.019)	(0.029)
Fraud(t) × Fraud Ind.	0.008	-0.019	-0.056***	-0.040
	(0.041)	(0.031)	(0.015)	(0.037)
Fraud(t+1) × Fraud Ind.	-0.056	-0.007	-0.055**	-0.081**
	(0.042)	(0.029)	(0.022)	(0.037)
Fraud(t+2) × Fraud Ind.	-0.041	-0.024	-0.072**	-0.075*
	(0.061)	(0.037)	(0.033)	(0.045)
Post(t+3) × Fraud Ind.	-0.044	-0.019	-0.049*	-0.031
	(0.044)	(0.030)	(0.029)	(0.046)
Post(t+4) × Fraud Ind.	-0.090*	-0.058**	-0.083***	-0.071
	(0.047)	(0.028)	(0.030)	(0.048)
Post(t+5) × Fraud Ind.	-0.108*	-0.090**	-0.103***	-0.091*
	(0.059)	(0.045)	(0.037)	(0.049)
Post(t+6) × Fraud Ind.	-0.122**	-0.026	-0.097**	-0.066
	(0.056)	(0.023)	(0.040)	(0.048)
Post(t+7) × Fraud Ind.	-0.110**	-0.021	-0.100**	-0.044
	(0.054)	(0.030)	(0.040)	(0.054)
Post(t+8) × Fraud Ind.	-0.161**	-0.035	-0.140**	-0.060
	(0.071)	(0.033)	(0.059)	(0.049)
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker
Observations	8,310,000	1,723,000	6,728,000	10,970,000
R-squared	0.613	0.606	0.608	0.629
Pre Fraud (t-1) PV	-0.520^	-0.229	-0.564***	-0.437^
	(-1.63)	(-1.23)	(-2.95)	(-1.55)
Pre Revelation (t+2) PV	-0.505**	-0.210^	-0.459***	-0.301^
	(-2.11)	(-1.55)	(-2.76)	(-1.33)

Table 6: Employee Mix

These panels report tests of employee mix using firm-wide tenure and differences in characteristics between exiting employees compared with new employees during the fraud, presented in Panel A and B, respectively. Panel A presents regression estimates of firm-wide employee tenure on the Fraud Indicator and a constant to show differences between the fraud firms and matched control firms, in the pre fraud, fraud, and post fraud periods in columns (1), (2), and (3), respectively. Panel B presents differences in employee characteristics and standard errors for the final full year of employment for leaving employees versus the first full year of employment for new employees. Leaving employees are the existing employees who left the firm in the first year of a fraud period. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

Panel A: Employee Tenure Mix in Event Time

	(1)	(2)	(3)
Fraud Period:	*Pre*	*Fraud*	*Post*
Dependent variable:			
Firm-wide Employee Tenure			
Fraud Indicator	-0.027**	-0.047***	-0.013
	(0.013)	(0.016)	(0.011)
Constant	0.316***	0.451***	0.540***
	(0.009)	(0.011)	(0.008)
Observations	1,300	800	1,600
R-squared	0.003	0.011	0.001

Panel B: Characteristics of Leaving Employees versus New Employees

	(1)	(2)
	T-Test of Differences Fraud Firm Employees Only (New minus Exiting)	
	Difference	Standard Error
Education	-0.237	(0.226)
Age	-0.563	(0.420)
Experience	-0.326	(0.359)
Annual Real Wages	-11,720***	(4,260)
Ln(*Annual Real Wages*)	-0.319***	(0.090)
Female	0.019	(0.022)
Observations	295,000	

Table 6: Employee Mix—continued

This panel reports estimates from OLS regression an analysis estimating equation (2): estimates for wage effects at fraud firms in the by-event-time years. The analysis uses a separate sample of newly hired employees in period $Fraud_t$ at the fraud or control firm. Standard errors are in parentheses and calculated with clustering by pre-fraud employer (i.e., fraud or control firm). Present value (PV) analyses are calculations of wage effects at the end of the indicated event-time period using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). For PVs, T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from the by-event-time year estimates. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

Panel C: New Employee Wages

	(1)
Dependent Variable = Ln(*Annual Real Wages*)	
Pre(t-3) × Fraud Ind.	0.009
	(0.015)
Pre(t-2) × Fraud Ind.	-0.025
	(0.015)
Pre(t-1) × Fraud Ind.	0.026
	(0.030)
Fraud(t) × Fraud Ind.	0.044
	(0.052)
Fraud(t+1) × Fraud Ind.	0.011
	(0.036)
Fraud(t+2) × Fraud Ind.	-0.006
	(0.029)
Post(t+3) × Fraud Ind.	-0.026
	(0.033)
Post(t+4) × Fraud Ind.	-0.061*
	(0.035)
Post(t+5) × Fraud Ind.	-0.081**
	(0.032)
Post(t+6) × Fraud Ind.	-0.070*
	(0.037)
Post(t+7) × Fraud Ind.	-0.057
	(0.046)
Post(t+8) × Fraud Ind.	-0.067
	(0.052)
Controls	Yes
Fixed Effects	Event-time, Year, Worker
Observations	3,306,000
R-squared	0.602
Pre Fraud (t-1) PV	-0.215
	(-1.19)
Pre Revelation (t+2) PV	-0.300*
	(-1.71)

Table 7. Employee Earnings and Market and Firm Heterogeneity

This table reports estimates from OLS regression analyses estimating equation (2): estimates for wage effects at fraud firms in the by-event-time years. In columns (1) and (2), the sample is divided into "thick" and "thin" markets that have above and below median, respectively, within-industry employees in the same county. In columns (3) and (4), the sample is divided into bankruptcy and non-bankruptcy fraud firms along with their matched pair. Standard errors are in parentheses and calculated with clustering by pre-fraud employer (i.e., fraud firm or matched control firm). Present value (PV) analyses are calculations of wage effects at the end of the indicated event-time period using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). For PVs, T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from the by-event-time year estimates. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. For PVs, significance at the 10% confidence level for a one-tailed test is indicated by ^. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

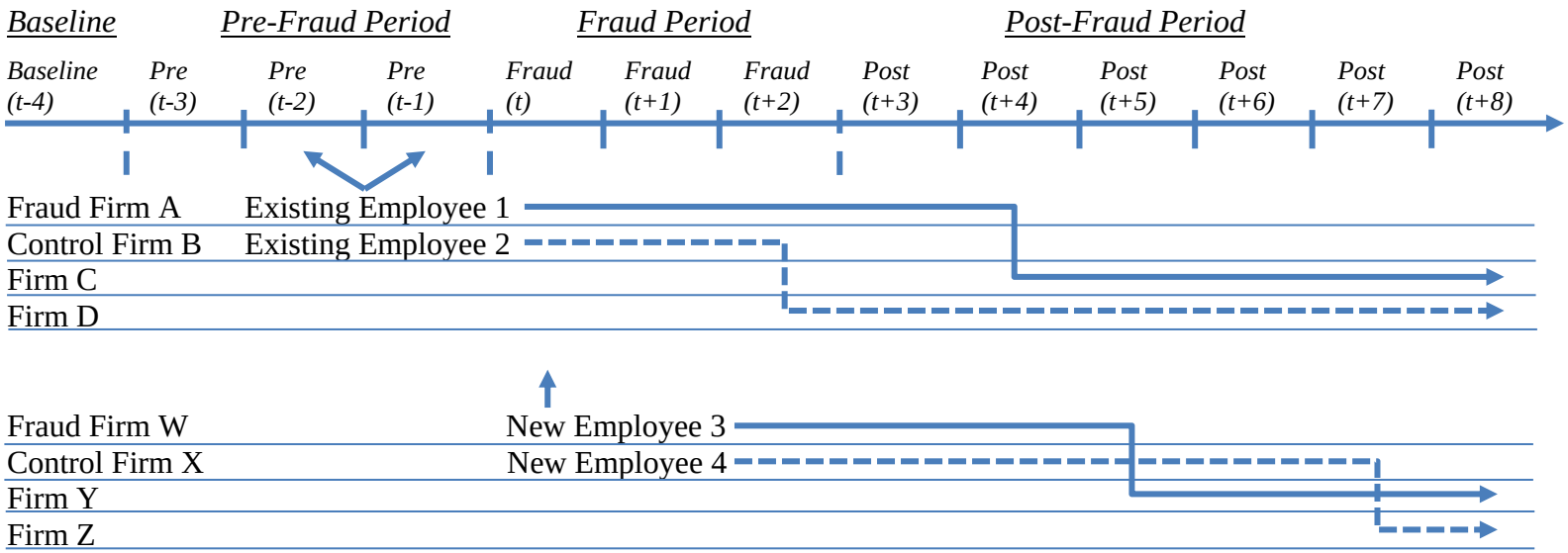
	(1)	(2)	(3)	(4)
Dependent Variable = Ln(*Annual Real Wages*)	Thick Markets	Thin Markets	Bankruptcy Fraud Firms	Non-Bankruptcy Fraud Firms
Pre(t-3) × *Fraud Ind.*	0.016	-0.029	-0.015	-0.013
	(0.021)	(0.031)	(0.045)	(0.026)
Pre(t-2) × *Fraud Ind.*	0.028	-0.044	-0.002	-0.014
	(0.034)	(0.030)	(0.050)	(0.031)
Pre(t-1) × *Fraud Ind.*	-0.012	-0.034	-0.030	-0.019
	(0.033)	(0.038)	(0.047)	(0.034)
Fraud(t) × *Fraud Ind.*	-0.018	-0.027	-0.088*	-0.017
	(0.038)	(0.024)	(0.046)	(0.028)
Fraud(t+1) × *Fraud Ind.*	0.042	-0.085**	-0.064	-0.030
	(0.042)	(0.042)	(0.059)	(0.040)
Fraud(t+2) × *Fraud Ind.*	-0.042	-0.090	-0.038	-0.061
	(0.036)	(0.055)	(0.056)	(0.047)
Post(t+3) × *Fraud Ind.*	-0.065	-0.063	-0.098	-0.054
	(0.044)	(0.043)	(0.065)	(0.043)
Post(t+4) × *Fraud Ind.*	-0.075	-0.116**	-0.164**	-0.083*
	(0.048)	(0.052)	(0.068)	(0.050)
Post(t+5) × *Fraud Ind.*	-0.098*	-0.146**	-0.255**	-0.104*
	(0.058)	(0.058)	(0.096)	(0.053)
Post(t+6) × *Fraud Ind.*	-0.094*	-0.161**	-0.222**	-0.122**
	(0.053)	(0.063)	(0.087)	(0.059)
Post(t+7) × *Fraud Ind.*	-0.056	-0.138**	-0.254***	-0.081
	(0.055)	(0.062)	(0.090)	(0.061)
Post(t+8) × *Fraud Ind.*	-0.017	-0.164**	-0.180**	-0.071
	(0.101)	(0.081)	(0.085)	(0.097)
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker
Observations	3,670,000	5,050,000	542,000	8,177,000
R-squared	0.602	0.606	0.614	0.605
Pre Fraud (t-1) PV	-0.313	-0.722**	-0.959**	-0.464^
	(-1.11)	(-2.43)	(-2.46)	(-1.52)
Pre Revelation (t+2) PV	-0.340^	-0.623**	-0.900***	-0.421^
	(-1.40)	(-2.47)	(-2.75)	(-1.61)

Table 8. Employee Movements and Pre-Fraud Wage Levels

This table reports estimates from OLS regression analyses estimating equation (2): estimates for wage effects at fraud firms in the by-event-time years. In column (1) (column (2)), the sample includes workers who leave (remain with) the fraud or control firm prior to or in (through at least) period *Post(t+5)*, i.e., leaves (stays) at the latest three years after the fraud concludes. In columns (3) and (4), we present subsamples of employees in the top 10% and bottom 90% of the pre-fraud wage distribution, respectively. Standard errors are in parentheses and calculated with clustering by pre-fraud employer (i.e., fraud or control firm). Present value (PV) analyses are calculations of wage effects at the end of the indicated event-time period using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). For PVs, T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from the by-event-time year estimates. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1)	(2)	(3)	(4)
Dependent Variable = Ln(*Annual Real Wages*)	Leavers	Stayers	Top 10%	Bottom 90%
Pre(t-3) × Fraud Ind.	-0.034	-0.004	-0.015	-0.012
	(0.025)	(0.030)	(0.030)	(0.025)
Pre(t-2) × Fraud Ind.	-0.045*	-0.018	0.008	-0.015
	(0.026)	(0.035)	(0.033)	(0.028)
Pre(t-1) × Fraud Ind.	-0.034	-0.045	0.047	-0.029
	(0.032)	(0.033)	(0.040)	(0.032)
Fraud(t) × Fraud Ind.	-0.016	-0.018	0.011	-0.028
	(0.034)	(0.024)	(0.034)	(0.027)
Fraud(t+1) × Fraud Ind.	-0.025	-0.018	0.002	-0.036
	(0.039)	(0.046)	(0.044)	(0.039)
Fraud(t+2) × Fraud Ind.	-0.086	-0.026	0.029	-0.078*
	(0.057)	(0.039)	(0.048)	(0.046)
Post(t+3) × Fraud Ind.	-0.037	-0.024	0.001	-0.065
	(0.054)	(0.041)	(0.043)	(0.042)
Post(t+4) × Fraud Ind.	-0.076	-0.024	-0.036	-0.098**
	(0.069)	(0.047)	(0.045)	(0.049)
Post(t+5) × Fraud Ind.	-0.132*	-0.054	-0.034	-0.128**
	(0.070)	(0.052)	(0.043)	(0.052)
Post(t+6) × Fraud Ind.	-0.134*	-0.083	-0.023	-0.144**
	(0.073)	(0.057)	(0.050)	(0.058)
Post(t+7) × Fraud Ind.	-0.141*	-0.039	-0.021	-0.107*
	(0.073)	(0.057)	(0.055)	(0.060)
Post(t+8) × Fraud Ind.	-0.187*	-0.018	-0.038	-0.099
	(0.096)	(0.085)	(0.078)	(0.098)
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker	Event-time, Year, Worker
Observations	4,633,000	4,087,000	870,000	7,850,000
R-squared	0.559	0.676	0.538	0.565
Pre Fraud (t-1) PV	-0.594*	-0.231	-0.071	-0.579*
	(-1.66)	(-0.80)	(-0.26)	(-1.95)
Pre Revelation (t+2) PV	-0.552*	-0.201	-0.125	-0.518**
	(-1.78)	(-0.81)	(-0.55)	(-2.05)

Figure 1: A Fraud Example, Timeline, and Employees



This figure is a representation of the accounting-fraud timeline. The fraud is split into three periods. The "Baseline" period is the first year prior to the three periods of interest, *Baseline*$_{t-4}$. The "Pre-Fraud Period" extends for up to three years prior to the beginning of the fraud from the Accounting and Auditing Enforcement Release (AAER). We indicate these years as *Pre(t-3)*, *Pre(t-2)*, and *Pre(t-1)*. The "Fraud Period" extends for the length of the fraud and must result in misreporting of an annual financial statement (e.g., a single quarter of fraud that is corrected within a fiscal year would be excluded). The Fraud Period is determined by the start year and end year of financial misrepresentation from the AAER. We indicate these years as *Fraud(t)*, *Fraud(t+1)*, and *Fraud(t+2)*. For long-lasting frauds, we normalize this period to a maximum of three years by indicating additional fraud years as *Fraud(t+2)*. The "Post-Fraud Period" extends for up to six years after the conclusions of the fraud from the AAER. We indicate these years as *Post(t+3)*, *Post(t+4)*, *Post(t+5)*, *Post(t+6)*, *Post(t+7)*, and *Post(t+8)*.

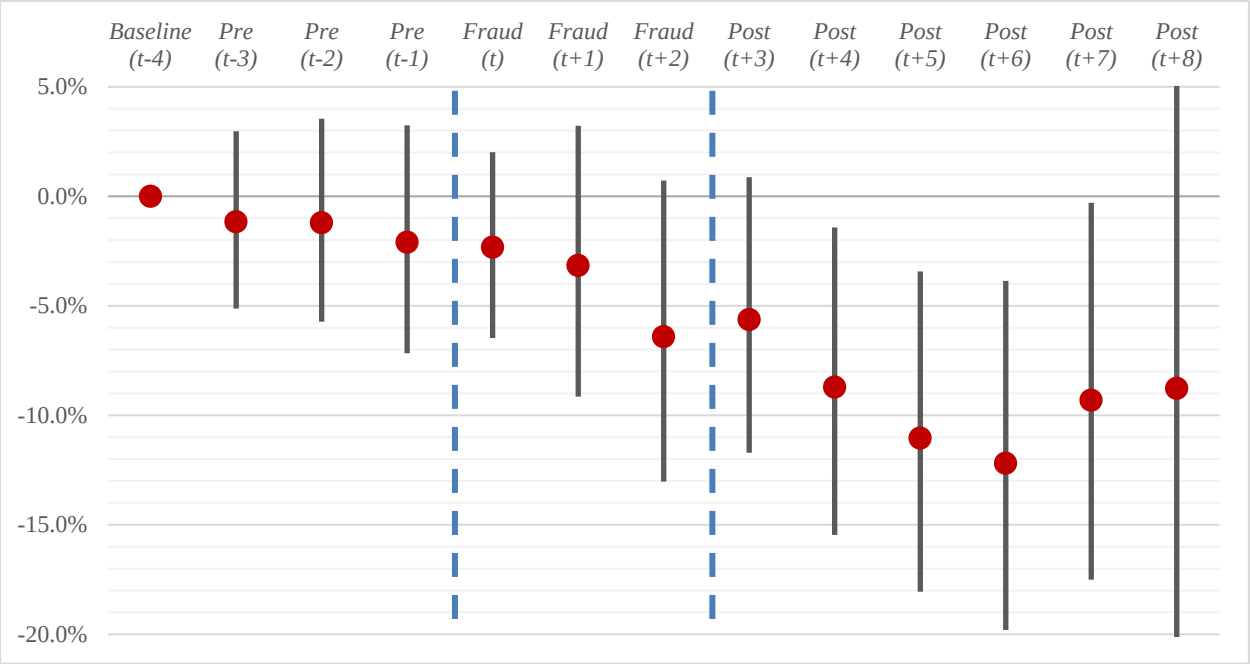
We classify employees into two types. "Existing Employees" are workers at fraud (or control) firms prior to the beginning of the fraud indicated in the AAER. We require that existing employees worked for a fraud firm or a control firm for the last two years before a fraud firm engaged in accounting fraud, *Pre(t-2)* and *Pre(t-1)*. We do not require that we are able to observe the hire date if the employee works for the firm before our sample begins. Existing employees comprise our main sample across most tables, i.e., all except Table 6. "New Employees" are workers at fraud (or control) firms hired during the Fraud Period. We require that new employees were hired in the first year of a fraud period by a fraud firm or a control firm, *Fraud(t)*. We report results for new employees in Table 6.

Our worker-year panel is not limited to the wage series at the treatment and control firms. We track employees as they move across firms. As an example, Existing Employee 1 might switch from Fraud Firm A to a third firm, Firm C, in event-time Post(t+4). We show other examples above, i.e., Existing Employee 2 moving from Control Firm B to Firm D in *Fraud(t+2)*, New Employee 3 moving from Fraud Firm W to Firm Y in Post(t+5), and New Employee 4 moving from Control Firm X to Firm Z in *Post(t+7)*, if Firms A, B, C, D, W, X, Y, and Z are in our sample states. We also have the wage series in the Baseline and Pre-Fraud Periods.

Figure 2: Dynamics of Earnings for Fraud Firm Employees

This figure shows magnitude estimates from OLS regression analyses estimating equation (2): estimates for wage effects at fraud firms in the by-event-time years. Point estimates are incremental earnings of employees at fraud firms relative to those at matched control firms. We adjust the coefficient estimates from column (3) in Table 3 to percentages. We also show 90% confidence interval estimates as vertical bars through the point estimates; standard errors are calculated with clustering by pre-fraud employer (i.e., fraud firm or matched control firm).

Panel A: Estimates from Table 3 Column (1): with Year Fixed Effects



Panel B: Estimates from Table 3 Column (3): with Year x Industry x County Fixed Effects

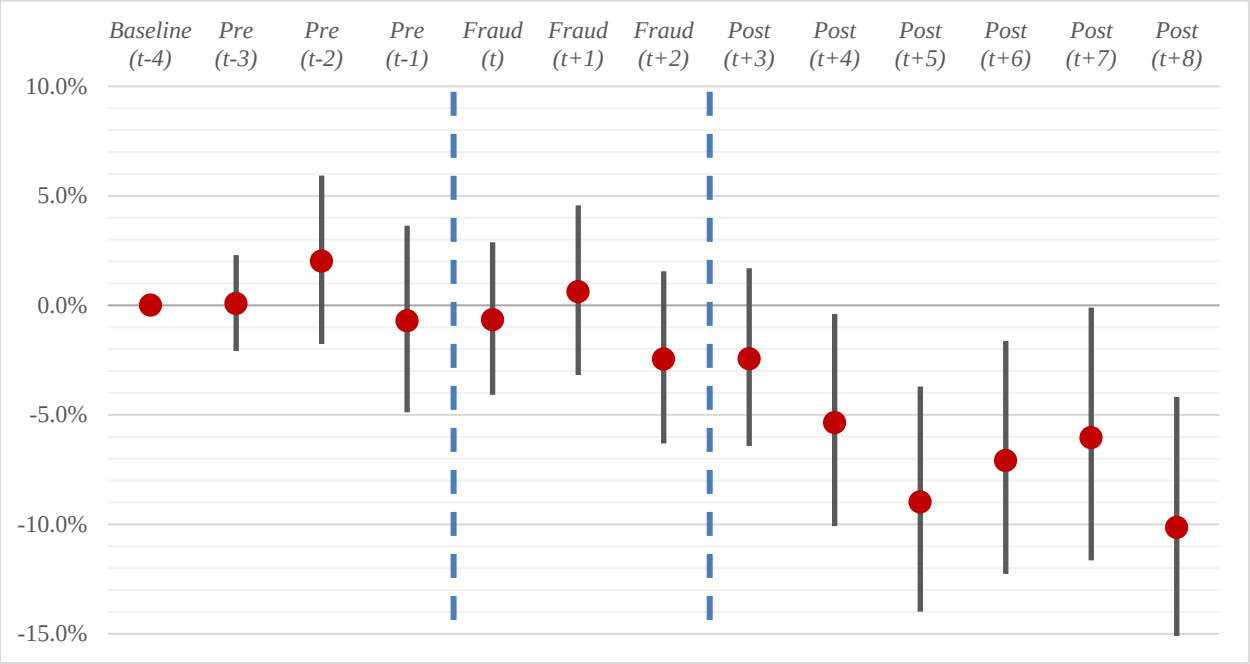
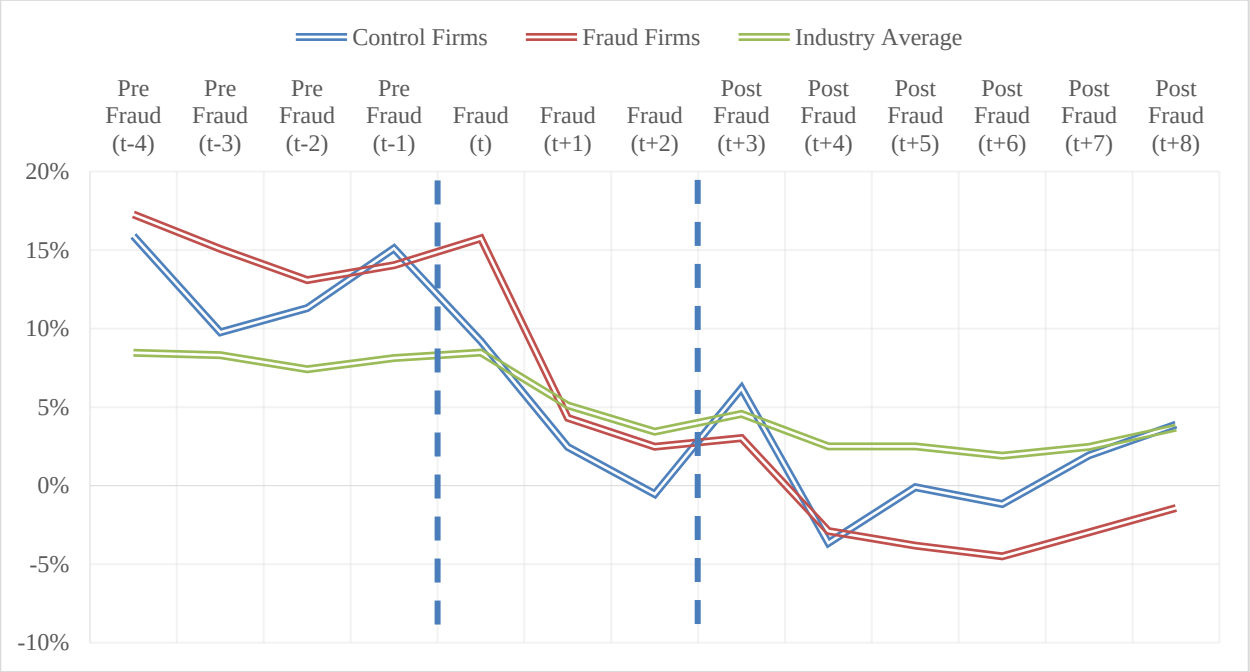


Figure 3: Employment Growth Levels

This figure shows estimates for employment growth at fraud firms in the by-event-time years. Point estimates are growth levels at fraud firms, matched control firms, and the industry average. In Panel A, we use LBD data. In Panel B, we use Compustat data.

Panel A: LBD Data



Panel B: Compustat Data

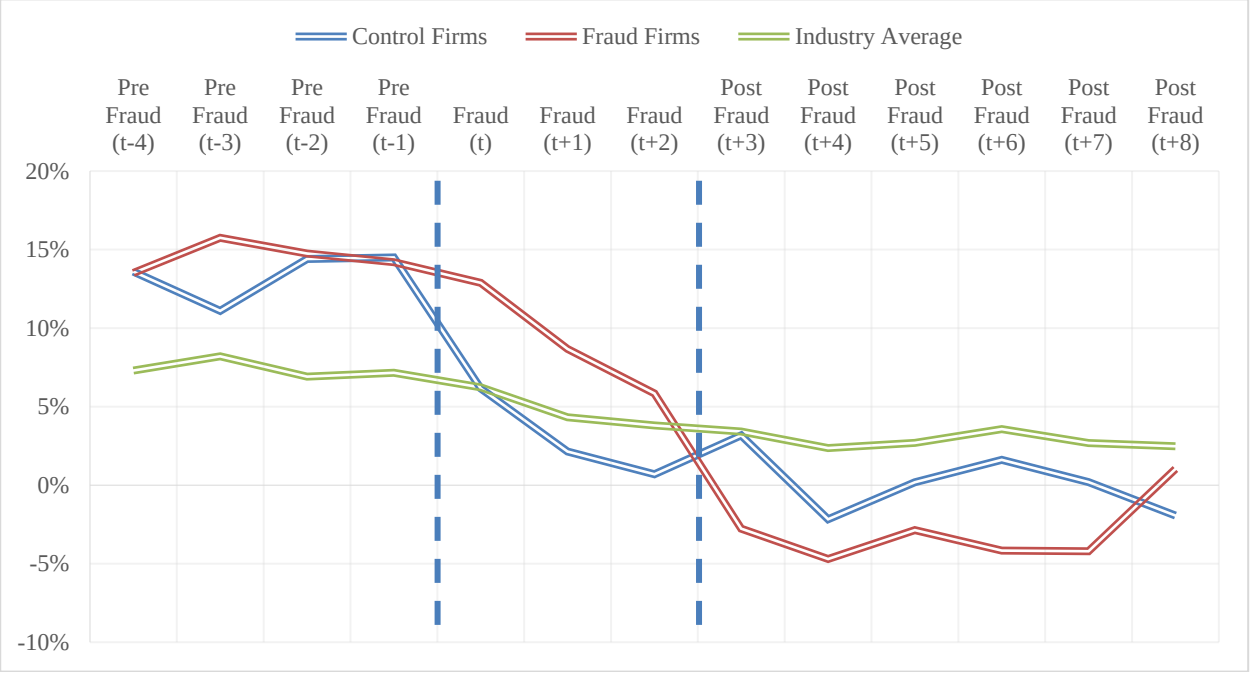
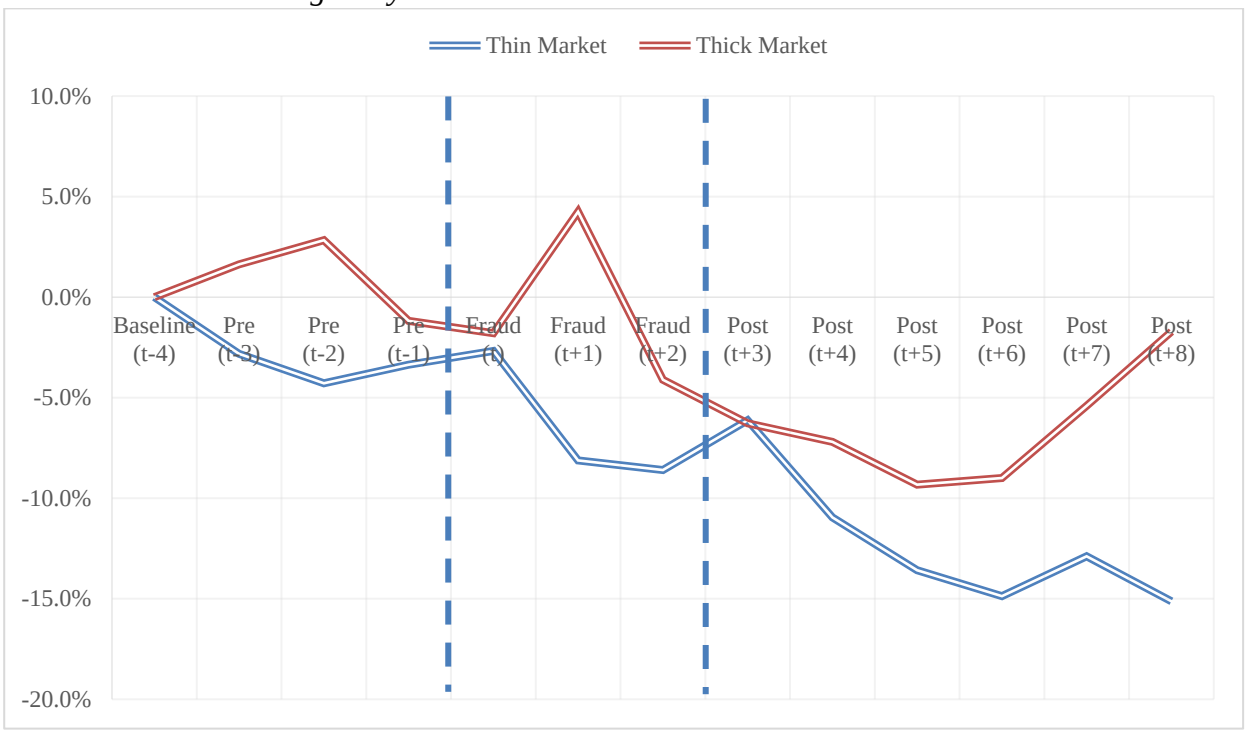


Figure 4: Wage Trends across Market and Firm Heterogeneity

This figure shows magnitude estimates from Table 7. We adjust the coefficient estimates to percentages. Wage trends are incremental for fraud firm employees relative to matched controls.

Panel A: Market Heterogeneity



Panel B: Firm Heterogeneity

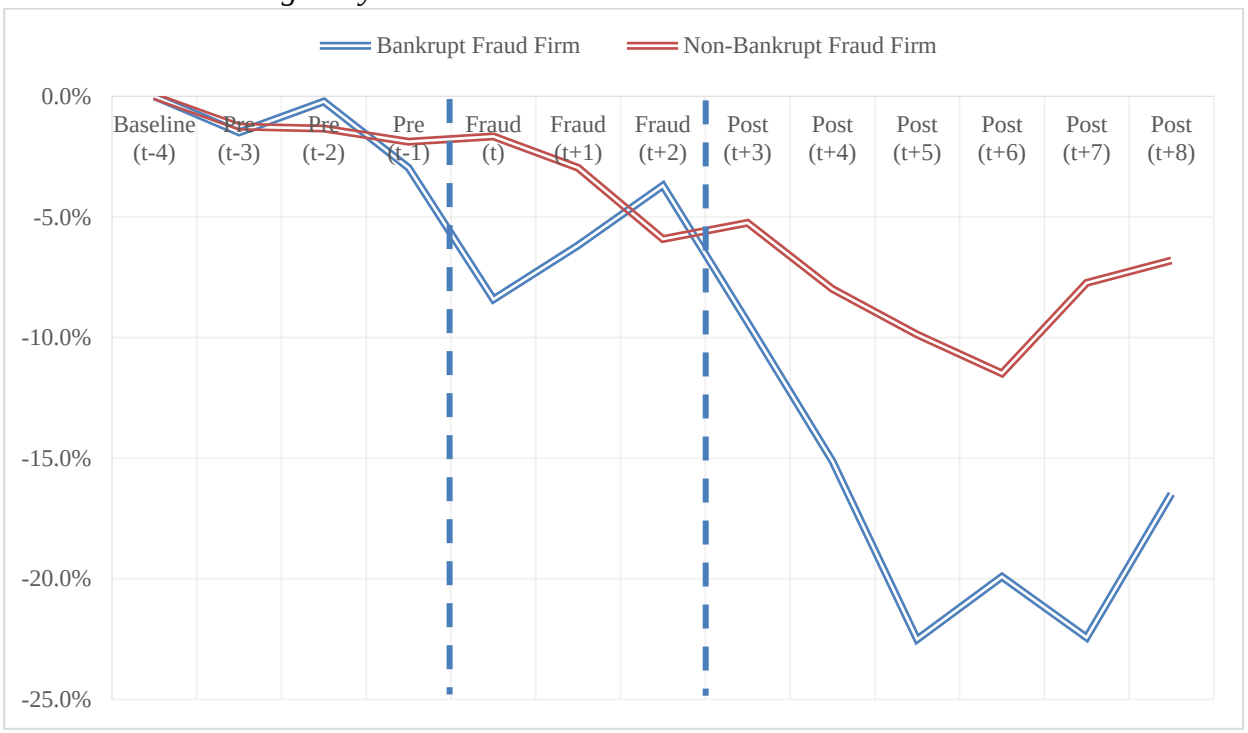
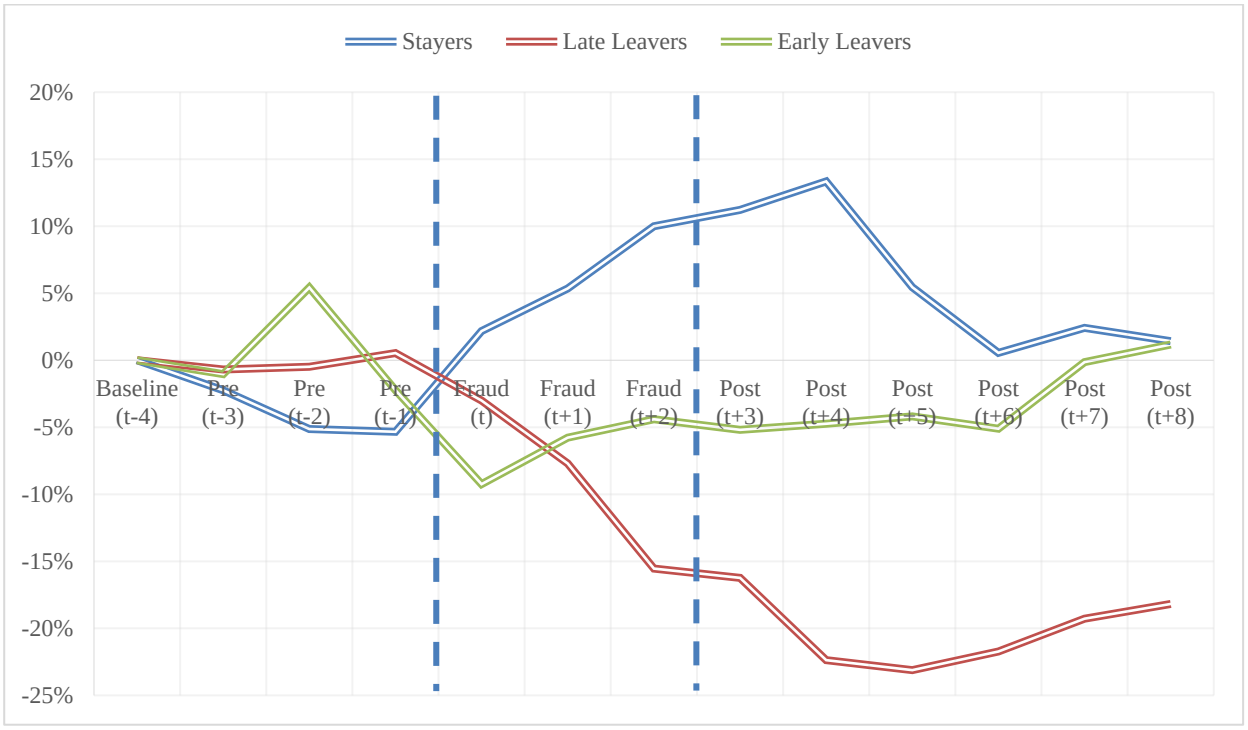


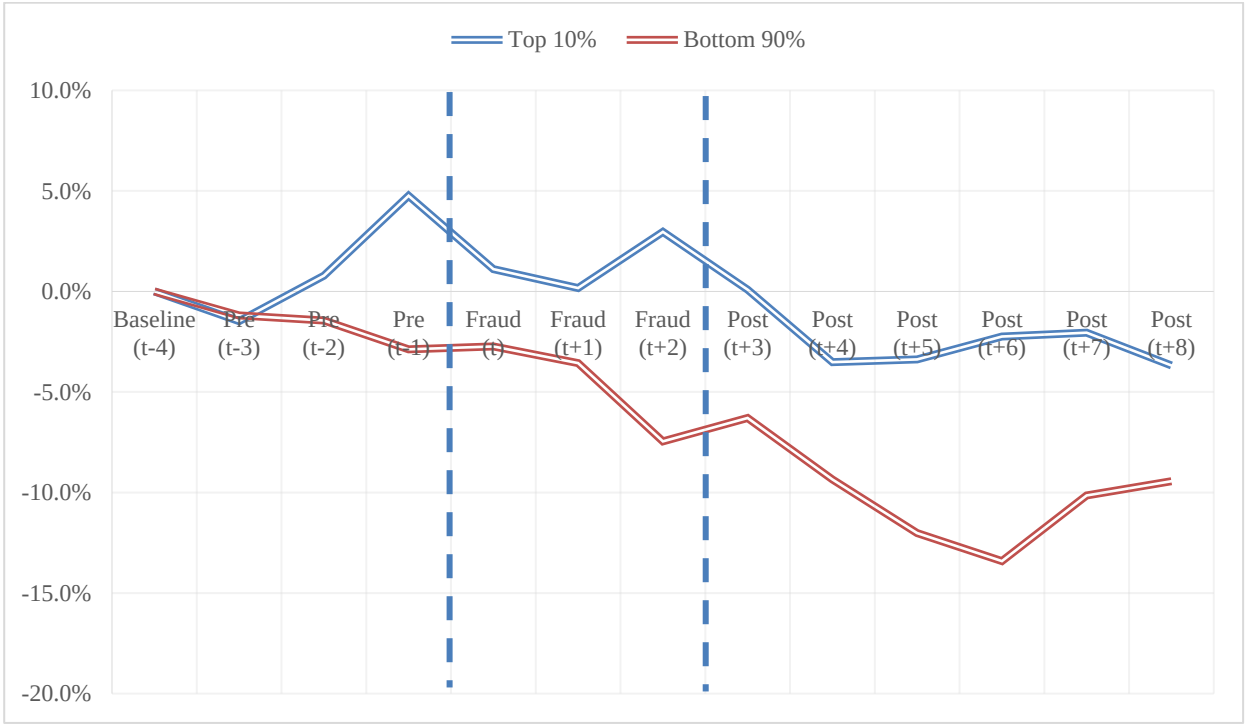
Figure 5: Wage Trends across Worker Heterogeneity

This figure shows magnitude estimates from an untabulated analysis for Panel A and Table 8 for Panel B. We adjust the coefficient estimates to percentages. Wage trends are incremental for fraud firm employees relative to matched controls.

Panel A: Worker Movements



Panel B: Pre-Fraud Wage Levels



Internet Appendix:

Fraudulent Financial Reporting and the Consequences for Employees

Jung Ho Choi

Brandon Gipper

- Table 1. Public Records for Fraud-Bankruptcy and Simple Bankruptcy Cases
- Table 2. Covariance Balance in Pre-Fraud Years
- Table 3. Qualitative Analysis of Bankruptcy Firms
- Table 4: During-Fraud Control Matches with Unadjusted Revenue Growth
- Table 5. Robustness excluding Financial Crisis Years
- Figure 1: Growth and Wages for Frauds of Varying Length
- Figure 2: Manuscript Table 4
- Figure 3: Manuscript Table 6 Panel C
- Figure 4: Firm-wide Tenure in Event Time
- Figure 5: Capital Expenditures in Event Time

Internet Appendix Table 1. Public Records for Fraud-Bankruptcy and Simple Bankruptcy Cases

This table presents fraud-bankruptcy and simple bankruptcy cases' public filings. The italics are added by the authors.

Panel A: Enron (Fraud-Bankruptcy Case)

	Description
Annual Report 1999	To Our Shareholders … We believe the future will be even more rewarding. We remain the world's leader in wholesale and retail energy services. Our new broadband subsidiary, Enron Broadband Service, is redefining Internet performance by designing and supplying a full range of premium broadband delivery services. The value of products bought and sold on our new eCommerce platform, EnronOnline™, is destined to exceed the value transacted on any current eCommerce web site. *To reap greater growth and value in our traditional energy businesses without a parallel increase in capital spending, we have evolved into a series of global networks – each of which is leader in tis specific region.* These networks work our physical assets harder and drive more high-return products and services into the market. We believe that our broad networks will give us unbeatable scale and scope in every business in every region in which we operate. … We are doing something that is recognized outside the company. This year, in addition to again naming us "America's Most Innovative Company," *Fortune ranked Enron "No.1 in Quality of Management" and "No.2 in Employee Talent" of all American companies.* The magazine also acknowledged us as one of the 25 best places to work in America. We recognize that our intellectual capital is our most important asset, and we cherish it. All employees are shareholders. Our values of communication, integrity, respect and excellence are equally applicable to our dealings with each other as with our customers and suppliers. …
AAER No. 1640	SEC CHARGES ANDREW S. FASTOW, FORMER CHIEF FINANCIAL OFFICER OF ENRON, WITH FRAUD *The Securities and Exchange Commission filed a civil enforcement action today against Andrew S. Fastow, the former chief financial officer of Enron Corp., alleging violations of the anti-fraud, periodic reporting, books and records, and internal controls provisions of the federal securities laws.* The Commission is seeking disgorgement of all ill-gotten gains, including all compensation received subsequent to the commencement of the alleged fraud, civil money penalties, a permanent bar from acting as a director or officer of a publicly held company, and an injunction from future violations of the federal securities laws. The Commission brought this action in coordination with the Justice Department's Enron Task Force, which filed a related criminal complaint against Fastow. ….. *Nigerian Barges: In December 1999, Enron and a financial institution entered into a sham "sale" transaction that enabled Enron to book approximately $12 million in earnings in 1999.* In the transaction, the financial institution agreed to "buy" from Enron an interest in certain power-producing barges in Nigeria based on an express oral

	promise from Fastow that Enron would arrange to take the financial institution out of the investment within six months. Enron also agreed to a specified profit for the financial institution's "investment." The transaction closed at the end of December 1999. Six months later, Fastow fulfilled his promise to take the financial institution out of the deal. He arranged for a partnership he controlled, LJM2 Co-Investment, L.P. ("LJM2") to purchase the financial institution's interest on the previously-agreed terms.

Panel B: Luckin Coffee (Fraud-Bankruptcy Case)

	Description
Form F-1 2019	Business / Our Strengths / Leading and Fastest Growing Player Driving Coffee Consumption in China We are China's second largest and fastest-growing coffee network, in terms of number of stores and cups of coffee sold, according to the Frost & Sullivan Report. *We started our business in October 2017 and within 18 months, we expanded from a single trial store in Beijing to 2,370 stores in 28 cities in China as of March 31, 2019.* In 2018, we sold close to 90 million coffee and other high quality product items. Since inception, we have successfully built a brand of distinguished value propositions—high quality, high affordability and high convenience. Our Luckin brand is one of the most recognized coffee brands in China, according to the Frost & Sullivan Report. *China coffee industry is in its early stage and has huge growth opportunities. According to the Frost & Sullivan Report, the average coffee consumption per capita in China was only 6.2 cups in 2018, compared to 867.4 cups in Germany, 388.3 cups in the United States, 279.0 cups in Japan, 249.5 cups in Hong Kong and 209.4 cups in Taiwan*. According to the Frost & Sullivan Report, the average coffee consumption per capita in China is expected to grow from 6.2 cups in 2018 to 10.8 cups in 2023, and the market size of China's freshly brewed coffee is expected to grow from RMB39.0 billion in 2018 to RMB157.9 billion in 2023, representing a CAGR of 32.3%. Inconsistent qualities, high prices and inconvenience are three key pain points hampering the growth of China's freshly brewed coffee market. We offer consumers coffee products with high quality, high affordability and high convenience to address these pain points. *As we attract new coffee consumers and encourage existing consumers to consume more coffee, we contribute to the growth of China's coffee industry.* According to the Frost & Sullivan Customer Survey, 82.4% of the participants indicated that they started to drink coffee more frequently since they became a customer of Luckin. We believe we are well-positioned to capture the future growth of China's coffee industry.
Litigation Release No. 24987	Securities and Exchange Commission v. Luckin Coffee Inc., No. 1:20-cv-10631 (S.D.N.Y. filed Dec. 16, 2020) *The Securities and Exchange Commission today charged China-based company Luckin Coffee Inc. with defrauding investors by materially misstating the company's revenue, expenses, and net operating loss in an effort to falsely appear to achieve rapid growth and increased profitability and to meet the company's earnings estimates.* Luckin,

	whose American Depositary Shares traded on the NASDAQ until July 13, 2020, has agreed to pay a $180 million penalty to resolve the charges. *The SEC's complaint alleges that, from at least April 2019 through January 2020, Luckin intentionally fabricated more than $300 million in retail sales by using related parties to create false sales transactions through three separate purchasing schemes*. According to the complaint, certain Luckin employees attempted to conceal the fraud by inflating the company's expenses by more than $190 million, creating a fake operations database, and altering accounting and bank records to reflect the false sales. *The complaint further alleges that the company intentionally and materially overstated its reported revenue and expenses, and materially understated its net loss, in its publicly disclosed financial statements in 2019.* For example, Luckin allegedly materially overstated its reported revenue by approximately 28% for the period ending June 30, 2019, and by 45% for the period ending September 30, 2019, in its publicly disclosed financial statements. The complaint alleges that during the period of the fraud, Luckin raised more than $864 million from debt and equity investors. After Luckin's misconduct was discovered in the course of the annual external audit of the company's financial statements, Luckin reported the matter to and cooperated with SEC staff, initiated an internal investigation, terminated certain personnel, and added internal accounting controls.

Panel C: JCPenney (Simple Bankruptcy Case)

	Description
Form 10-K 2018	Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations … **<u>Current Initiatives</u>** *With the development of a new senior leadership team, management is in the process of reassessing strategies and evaluating the transformational needs of the business.* Going through this process, management's immediate priority is to reestablish the fundamentals of retail based on the following initiatives: • Reducing and enhancing our inventory position; • Strengthening our integrated omnichannel strategy; • Redesigning and improving core store processes; • Improving our shrink results; and • Revamping our merchandise assortments and strategies. First, we will continue our efforts to reduce and enhance our inventory position. For 2018, inventory was reduced by over 13%. For 2019, we will continue to clear unproductive inventory swiftly and thoughtfully and implement new processes and training across our organization to ensure proper inventory management and merchandising and prevent future build-up of excess inventory.

	Second, we will continue to strengthen our integrated omnichannel strategy to ensure we return to growth in our digital channels, but in a sustainable and profitable way. In 2019, we continue to focus on improving the overall customer experience from a digital standpoint and on strengthening our omnichannel retail capabilities to ensure we have a consistent experience regardless of whether our customer is shopping on-line, mobile or in-store. Third, we will redesign and improve core store processes while implementing enhanced technology tools to better equip our store associates to deliver on our customers' expectations. In 2019, we will have a critical focus on improving the productivity of our labor hours by delivering service to the customer where and when it matters most. Fourth, we are taking immediate action to improve our shrink results. We have made and will continue to make technology investments and staffing adjustments to improve our inventory shrink results. *Lastly, we will continue to revamp and rethink our merchandise assortments and strategies. In 2019, we made the decision to discontinue selling major appliances in an effort to improve financial performance and refocus on our legacy strengths in apparel and soft home as well as profitable growth opportunities.* We will also continue to leverage our best-in-class global sourcing and design organization across our product spectrum to enhance the style and quality we deliver, but also to improve profitability within our assortment.

Internet Appendix Table 2. Covariance Balance in Pre-Fraud Years

This table compares fraud firms' to control firms' characteristics in the years prior to accounting fraud, *Pre(t-4)*, *Pre(t-3)*, and *Pre(t-2)* in Panels A, B, and C, respectively. Accounting fraud and control firms are described in the manuscript. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Significance below these conventional levels is indicated with "ns." Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1) Fraud Firms	(2) Control Firms	(3) P-value Indicators
Panel A: *Pre(t-4)*			
Size	5.977	5.768	ns
Assets ($M)	4,772	2,507	*
Sales Growth	0.194	0.160	ns
Return on Assets	0.083	0.099	ns
Leverage	0.238	0.198	ns
Tobin's Q	2.094	2.425	ns
Employee growth	0.173	0.159	ns
Avg. Annual Real Wages ($Th)	38.32	40.06	ns
Employees (Headcount in U.S.)	9,506	9,628	ns
Panel B: *Pre(t-3)*			
Size	6.134	5.872	ns
Assets ($M)	4,887	2,856	ns
Sales Growth	0.2112	0.194	ns
Return on Assets	0.070	0.077	ns
Leverage	0.228	0.204	ns
Tobin's Q	2.013	2.416	*
Employee growth	0.151	0.098	ns
Avg. Annual Real Wages ($Th)	44.32	45.74	ns
Employees (Headcount in U.S.)	9,726	9,320	ns
Panel C: *Pre(t-2)*			
Size	6.216	6.069	ns
Assets ($M)	4,935	3,079	ns
Sales Growth	0.224	0.171	*
Return on Assets	0.069	0.080	ns
Leverage	0.248	0.216	ns
Tobin's Q	2.11	2.310	ns
Employee growth	0.131	0.113	ns
Avg. Annual Real Wages ($Th)	45.71	46.83	ns
Employees (Headcount in U.S.)	10,198	8,762	ns

Internet Appendix Table 3. Qualitative Analysis of Bankruptcy Firms

This table reports qualitative output from OLS regression analyses estimating a modified specification similar to equation (2) from the manuscript. Qualitative output includes coefficient signs and significance at conventional levels. The regression estimates wage effects at bankruptcy firms as a dependent variable on the by-bankruptcy-event-time year indicators from four years prior to the bankruptcy (the baseline year) through to six years after the bankruptcy (*Pre Bankrupt(t-3)* through *Post Bankrupt(t+6)*) interacted with a bankrupt firm indicator *Bankruptcy Ind.*. Other main effects and controls are also included. These analyses are similar to Graham et al. (2019). At the bottom, we give estimates for present value calculations of wage effects at the end of the indicated event-time period from all subsequent by-event-time year estimates using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from the by-event-time year estimates. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Significance below these conventional levels is indicated with "ns." Descriptive statistics, coefficient estimates, t-statistics, number of observations, and R-squared are not reported due to overlap with other publicly available analyses from the U.S. Census Bureau, so output is limited to qualitative reporting of coefficients to comply with disclosure requirements.

Dependent Variable = Ln(*Annual Real Wages*)	(1) Bankrupt Firms	(2) Bankrupt Firms with Fraud	(3) Bankrupt Firms without Fraud
Pre Bankrupt(t-3) × *Bankruptcy Ind.*	- ns	- ns	- ns
Pre Bankrupt(t-2) × *Bankruptcy Ind.*	- ns	-***	- ns
Pre Bankrupt(t-1) × *Bankruptcy Ind.*	- ns	-***	- ns
Post Bankrupt(t) × *Bankruptcy Ind.*	-***	- ns	-***
Post Bankrupt(t+1) × *Bankruptcy Ind.*	-***	-***	-***
Post Bankrupt(t+2) × *Bankruptcy Ind.*	-***	-***	-***
Post Bankrupt(t+3) × *Bankruptcy Ind.*	-***	-***	-***
Post Bankrupt(t+4) × *Bankruptcy Ind.*	-***	-***	-***
Post Bankrupt(t+5) × *Bankruptcy Ind.*	-***	-***	-***
Post Bankrupt(t+6) × *Bankruptcy Ind.*	-***	-***	-***
Controls and main effects	Yes	Yes	Yes
Fixed Effects	Year, Worker	Year, Worker	Year, Worker
Pre Bankruptcy (t-1) Present Value	-0.692*** (-4.62)	-1.035*** (-7.10)	-0.644*** (-4.03)

Internet Appendix Table 4: During-Fraud Control Matches with Unadjusted Revenue Growth

Panel A: Probit Model

This table shows the results of a probit model estimating a propensity score to engage in accounting fraud. This probit model corresponds to matching for the control firms in manuscript Table 5 Panel D for column (1). Accounting-fraud firms are identified by the AAER. Fraud firms are included in sample firms in the year prior to accounting fraud, *Pre(t-1)*. Non-fraud firms are included in sample firms if they operate businesses in the same industry as one of fraud firms in the year prior to accounting fraud. The sample period is from 1989 to 2008. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1)
Dependent Variable: *Fraud-Firm Indicator*	
Size	0.068***
	(0.017)
Unmanaged, Fraud Period Sales Growth	0.489***
	(0.111)
Return on Assets	0.437*
	(0.229)
Leverage	0.045
	(0.125)
Tobin's Q	0.018**
	(0.008)
Employee Growth	0.096
	(0.076)
Ln*(Avg. Annual Real Wages)*	-0.010
	(0.056)
Observations	26,500
Chi-squared	54.1
Pseudo R-squared	0.033

Internet Appendix Table 4: During-Fraud Control Matches with Unadjusted Revenue Growth—continued

Panel B. Comparison of Sample Fraud and Matched Control Firms

This table compares fraud firms' to control firms' characteristics in the year prior to accounting fraud, *Pre(t-1)*. These descriptive characteristics corresponds to matching for the control firms in manuscript Table 5 Panel D for column (1). Accounting-fraud firms are identified by the AAER. Control firms are matched to fraud firms based on a propensity score estimated in Panel A of Internet Appendix Table 4. The sample period is from 1989 to 2008. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau. ^ indicates that we did not extract these statistics for public disclosure.

	(1) Fraud Firms	(2) Control Firms	(3) Differences (Fraud minus Control)	
			Difference	Standard Error
Size	6.485	6.361	0.124	(0.241)
Assets ($M)	5,253	4,039	1,214	(1,362)
Sales Growth	0.199	0.199	-0.000	(0.039)
Return on Assets	0.077	0.070	0.006	(0.019)
Leverage	0.252	0.226	0.027	(0.029)
Tobin's Q	2.417	2.762	-0.345	(0.384)
Employee growth	0.129	0.197	-0.067	(0.045)
Avg. Annual Real Wages ($Th)	52.71	47.15	5.56^	^
Employees (Headcount in U.S.)	10,560	12,330	-1,770^	^
Observations	150	150		

Internet Appendix Table 4: During-Fraud Control Matches with Unadjusted Revenue Growth—continued

This table shows differences for averages of employees at fraud and control firms. These descriptive characteristics corresponds to matching for the control firms in manuscript Table 5 Panel D for column (1). Accounting-fraud firms in the sample commit financial misrepresentation from 1989 to 2008 according to the AAER. Control firms are matched to fraud firms based on a propensity score estimated in Panel A of Internet Appendix Table 4. Appendix Table A defines variables. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1) Fraud Firms	(2) Control Firms	(3) Differences (Fraud minus Control)	(4)
			Difference	Standard Error
Education	14.19	13.76	0.43	(0.280)
Age	37.90	36.52	1.38*	(0.836)
Experience	17.71	16.76	0.95	(0.668)
Annual Real Wages	71,370	52,870	18,500**	(7,915)
Female	0.452	0.485	-0.033	(0.058)
Observations	345,000	381,000		

Internet Appendix Table 5. Robustness excluding Financial Crisis Years

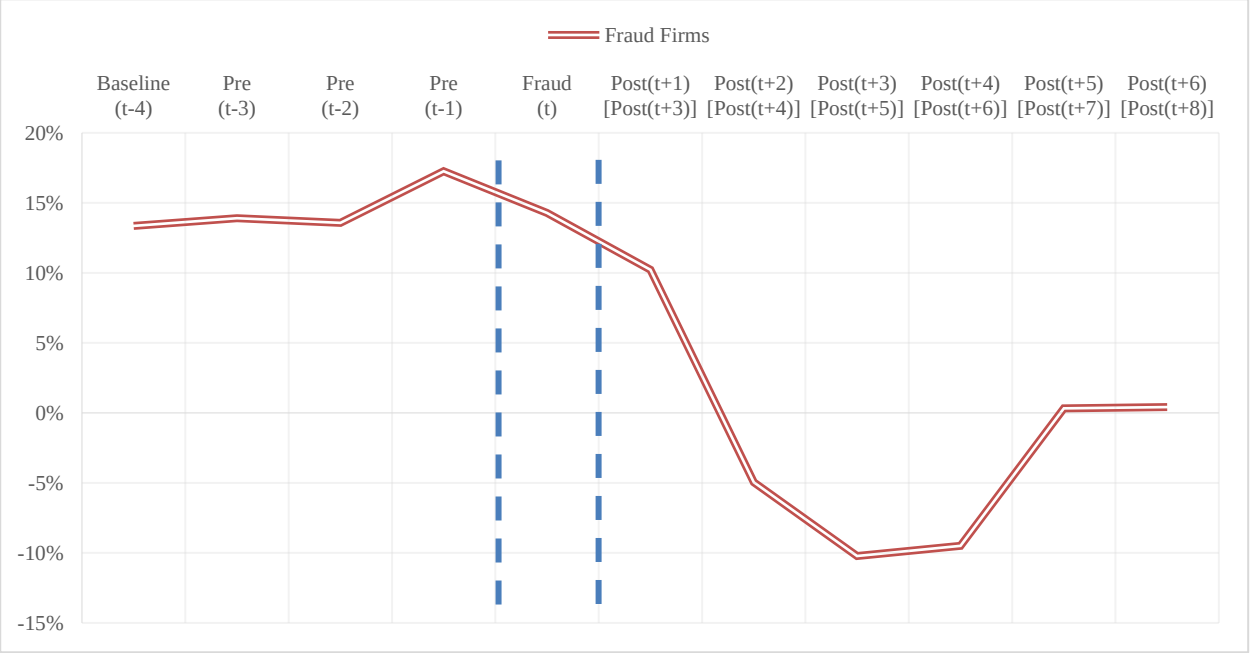
This table reports estimates for present value calculations of wage effects at the end of the indicated event-time period using inflation-adjusted discount rates from investment grade corporate bonds during our sample period (i.e., a discount rate of 4.47%). T-statistics are in parentheses and calculated with the delta method using the variance-covariance matrix from estimates for wage effects at fraud firms in the by-event-time years. Statistical significance at the 10%, 5%, and 1% confidence levels is indicated by *, **, and ***, respectively. For PVs, significance at the 10% confidence level for a one-tailed test is indicated by ^. Number of observations are rounded to comply with disclosure requirements of the U.S. Census Bureau.

	(1)	(2)
Controls and main effects	Yes	Yes
Fixed Effects	Year, Worker	Year × Industry, Worker
Pre Fraud (t-1) Present Value	-0.359	-0.292^
	(-1.15)	(-1.50)
Pre Revelation (t+2) Present Value	-0.336	-0.324**
	(-1.28)	(-2.02)

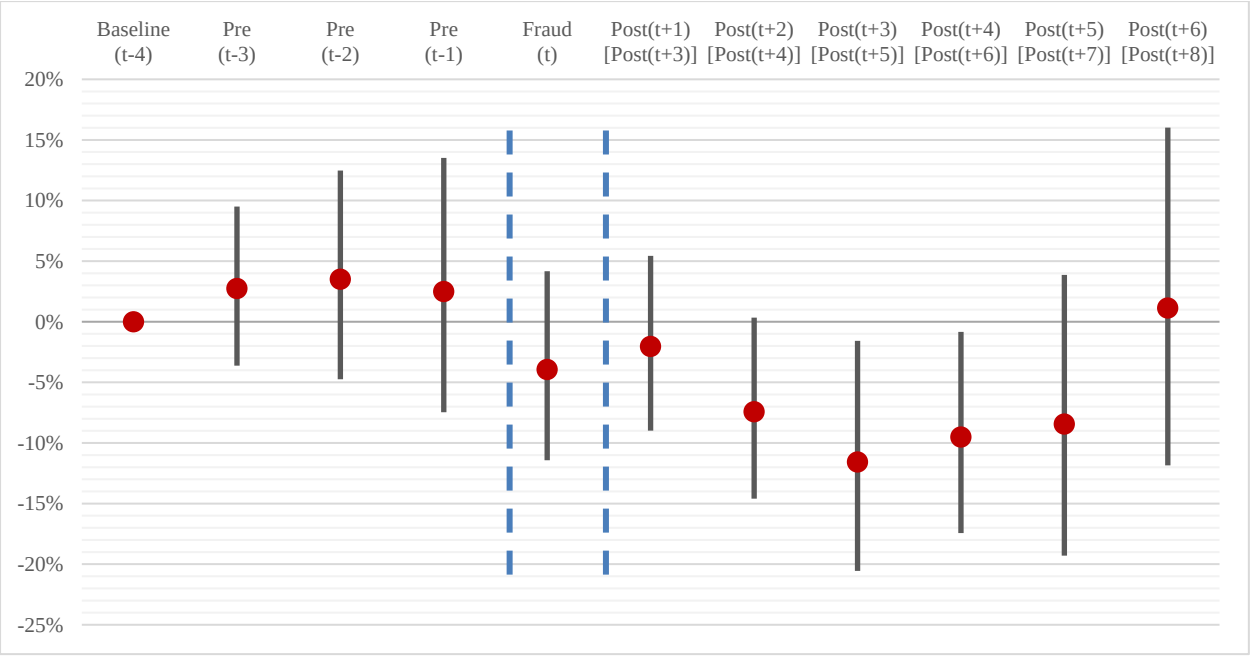
Internet Appendix Figure 1: Growth and Wages for Frauds of Varying Length

This figure shows estimates for employment growth and wage effects at fraud firms in the by-event-time years. Figure construction follows similar figures from the manuscript, i.e., Figures 2 and 3 for employment growth and wage effects, respectively. For one- and two-year frauds, we relabel the post period indicators to reflect the actual event timing and for ease of interpretation; actual variable names for the regression analyses are shown below in brackets.

Panel A: One Year Fraud Employment Growth

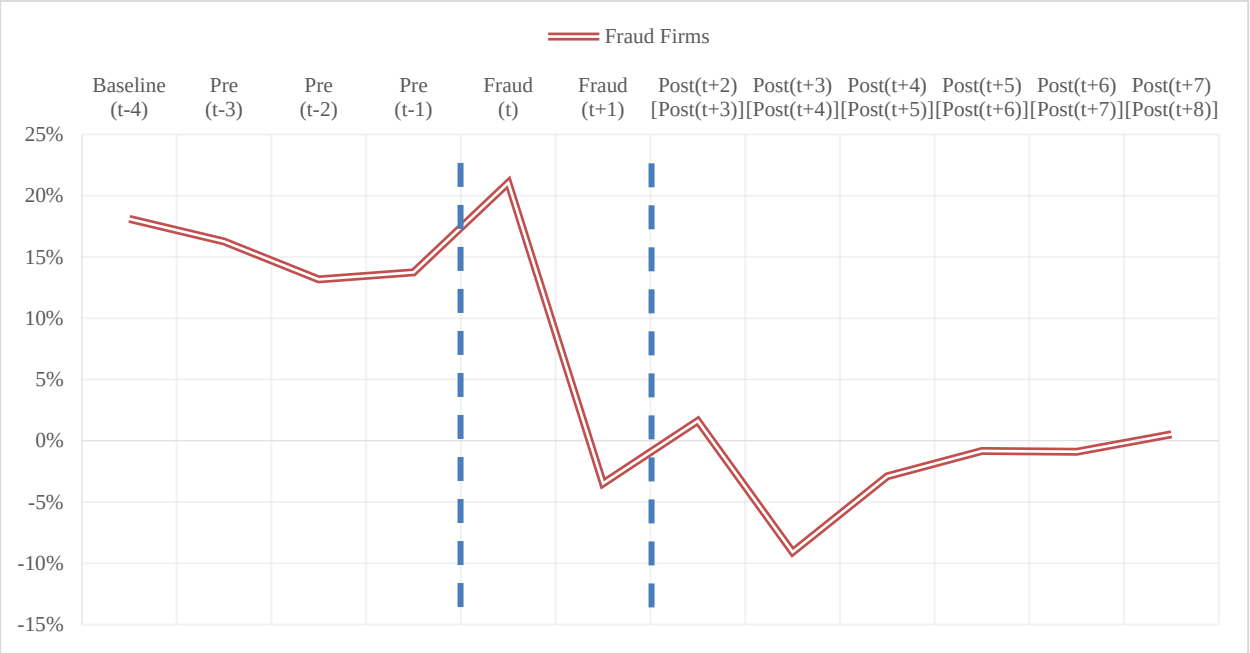


Panel B: One Year Fraud Wage Effects

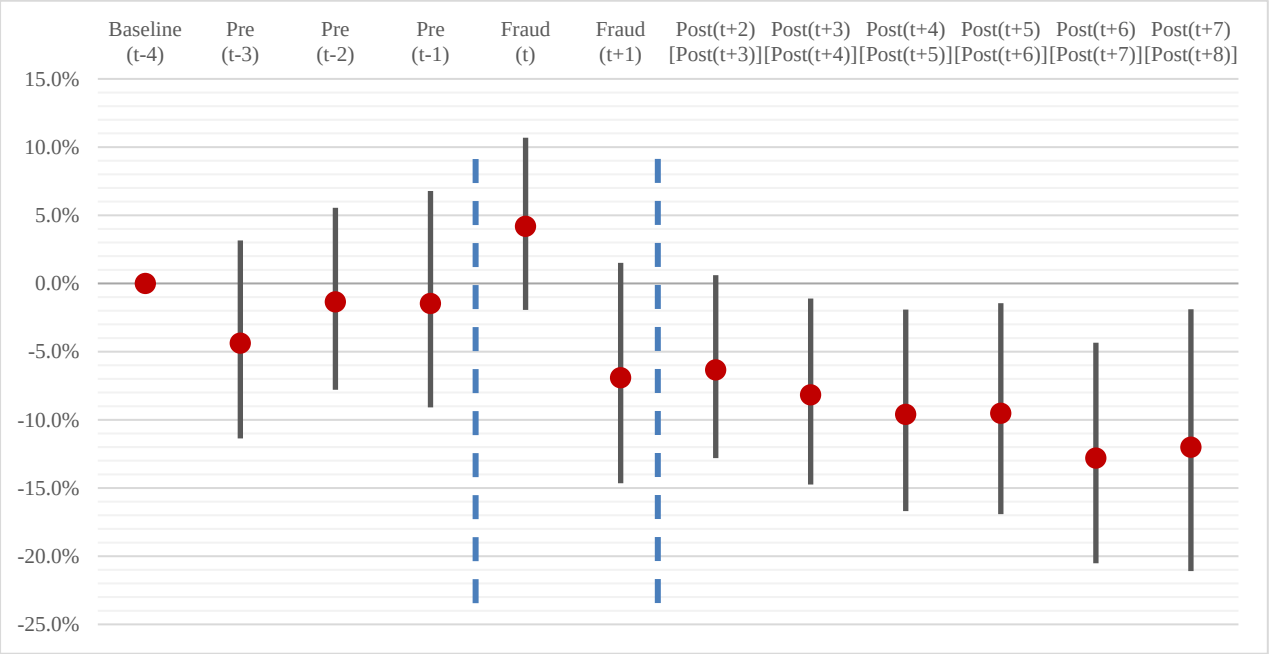


Internet Appendix Figure 1: Growth and Wages for Frauds of Varying Length—continued

Panel C: Two Year Fraud Employment Growth

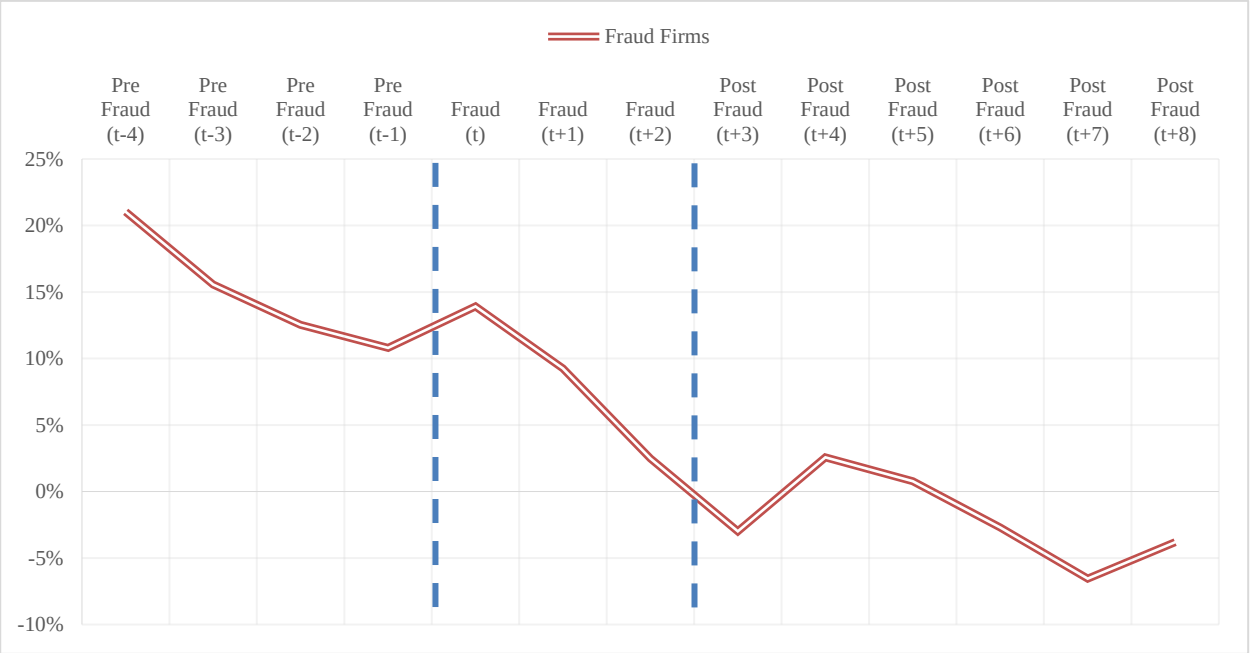


Panel D: Two Year Fraud Wage Effects

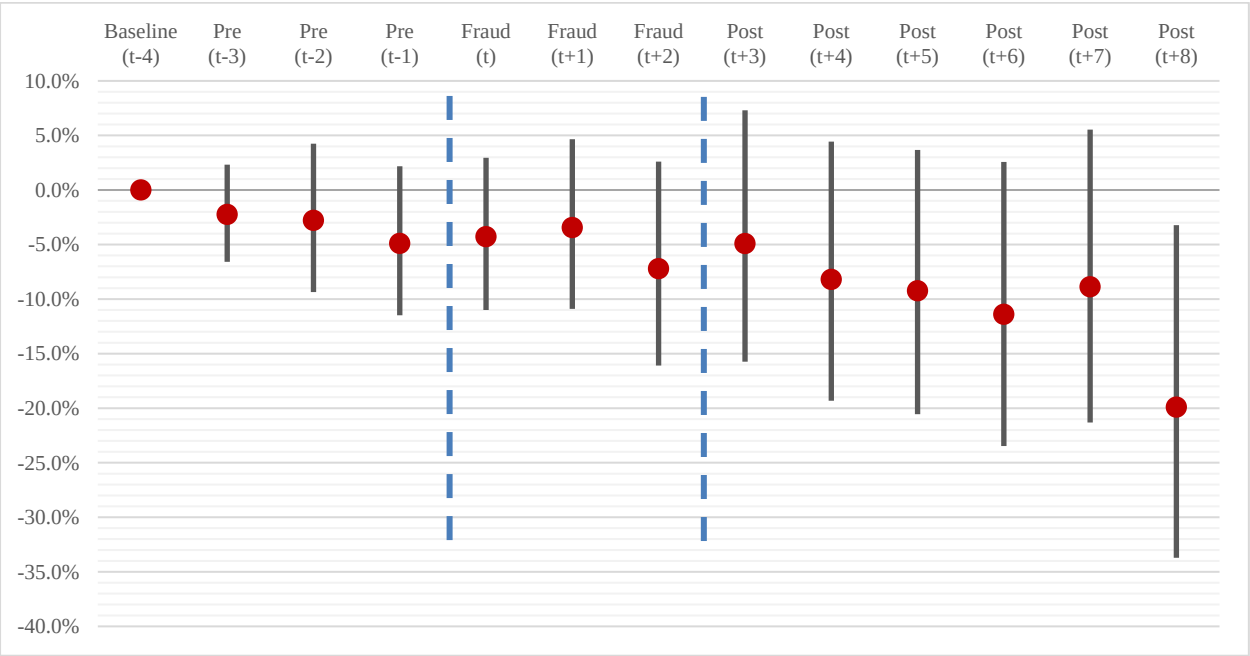


Internet Appendix Figure 1: Growth and Wages for Frauds of Varying Length—continued

Panel E: Three (and More) Year Fraud Employment Growth



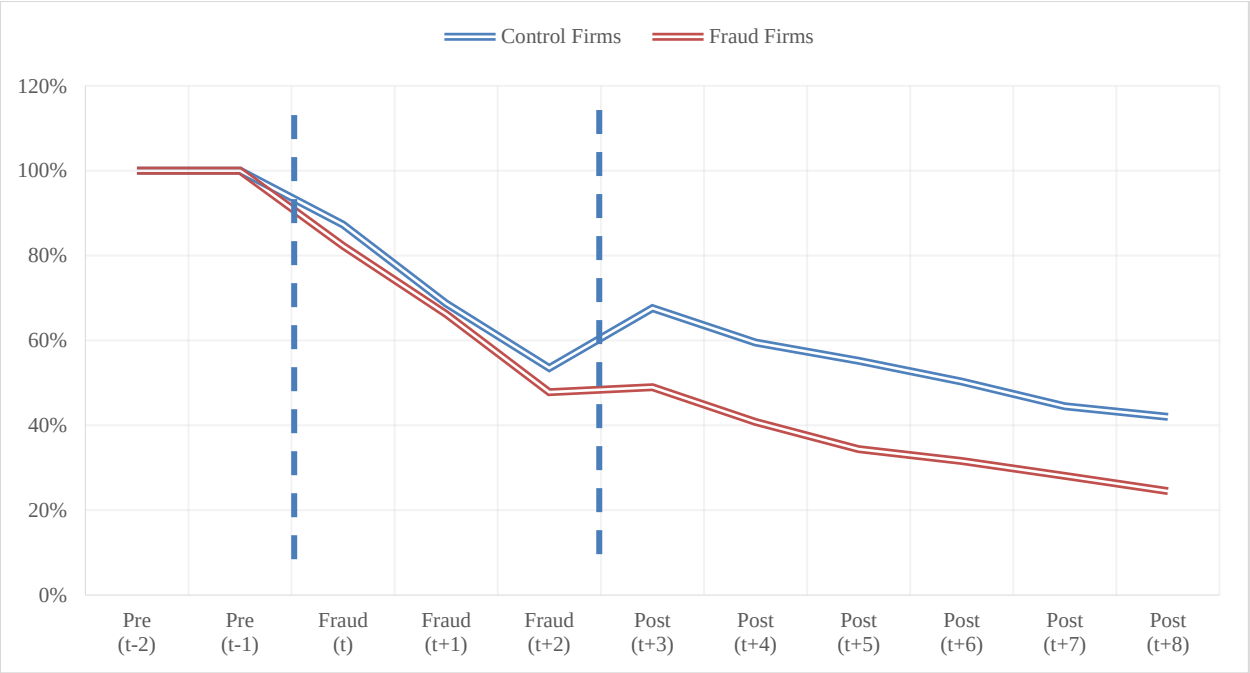
Panel F: Three (and More) Year Fraud Wage Effects



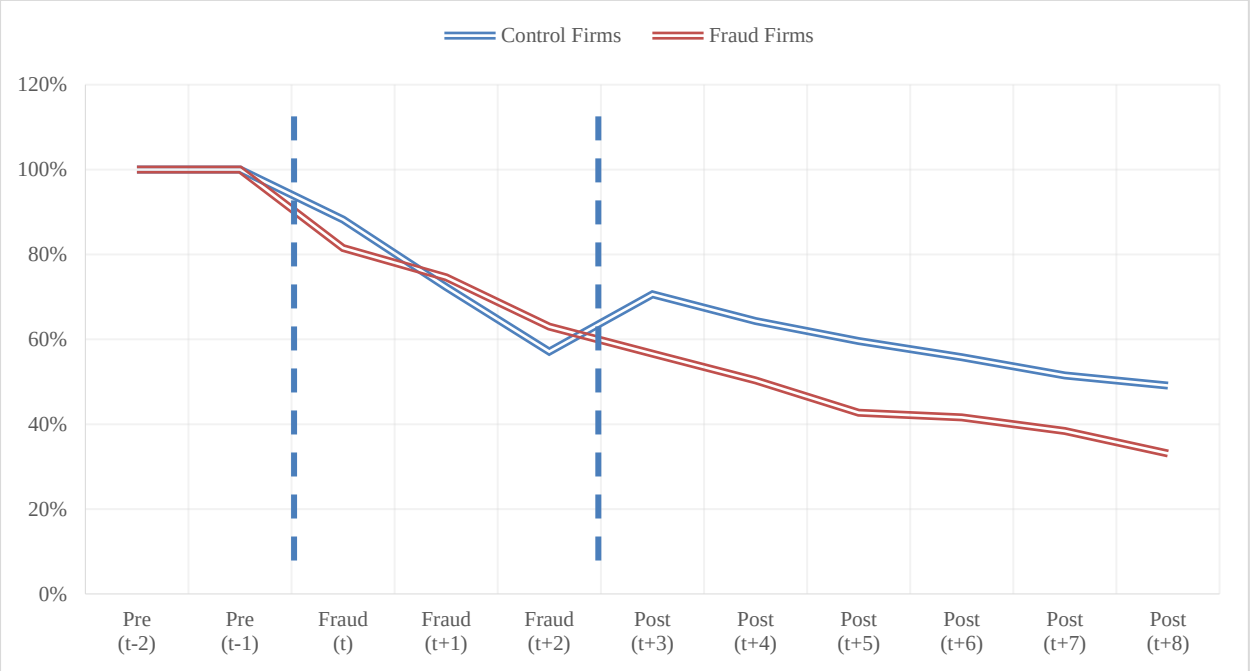
Internet Appendix Figure 2: Manuscript Table 4

This figure shows data from Table 4 in the manuscript. Table 4 shows averages and differences of employee retention at fraud and matched control firms.

Panel A: Percent Employees Retained at the Firm

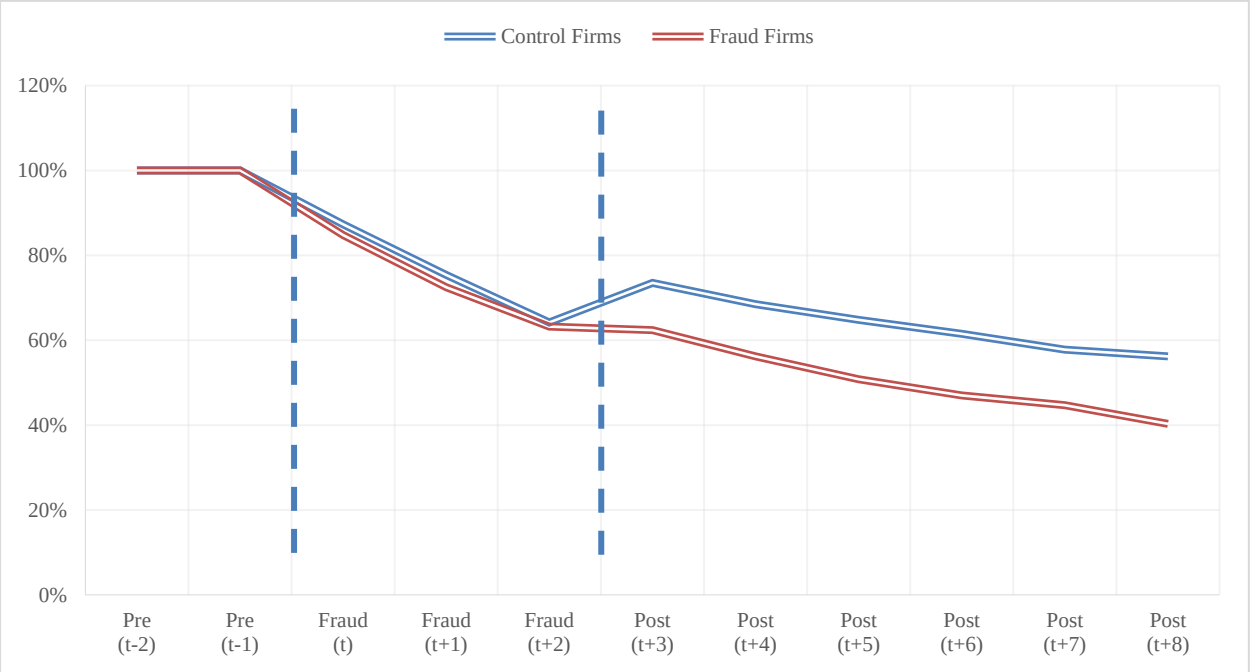


Panel B: Percent Employees Remaining within the Industry



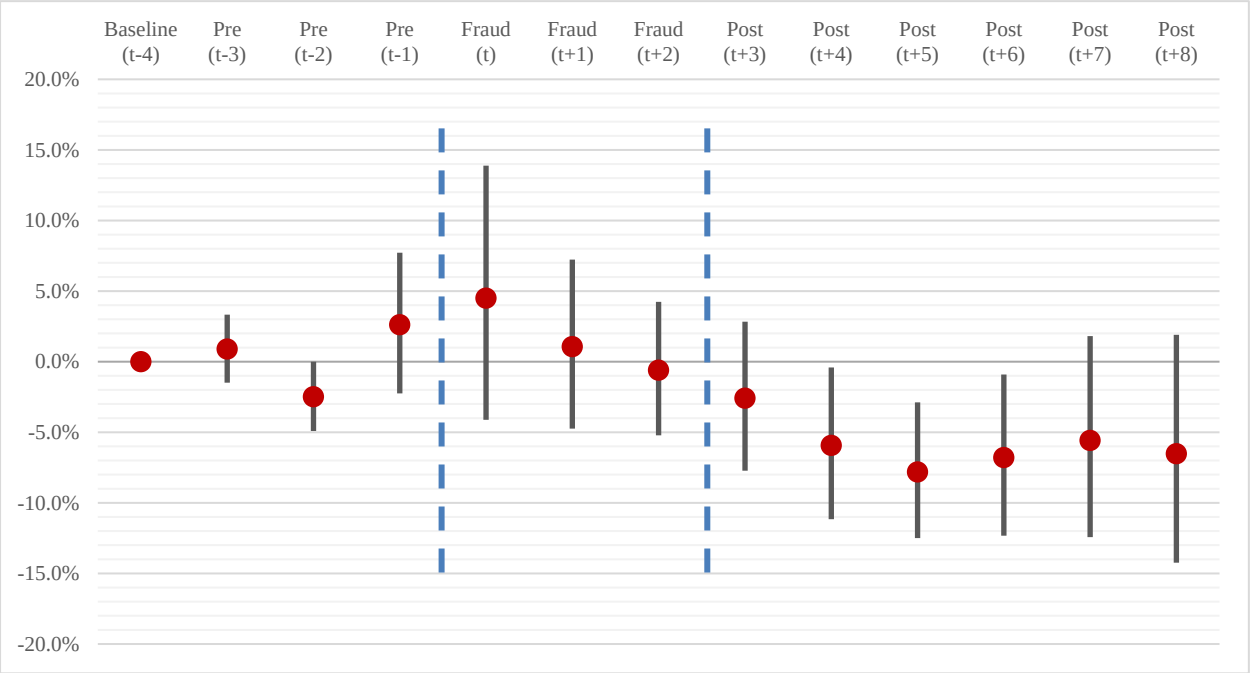
Internet Appendix Figure 2: Manuscript Table 4—continued

Panel C: Percent Employees Continuing Work within the County



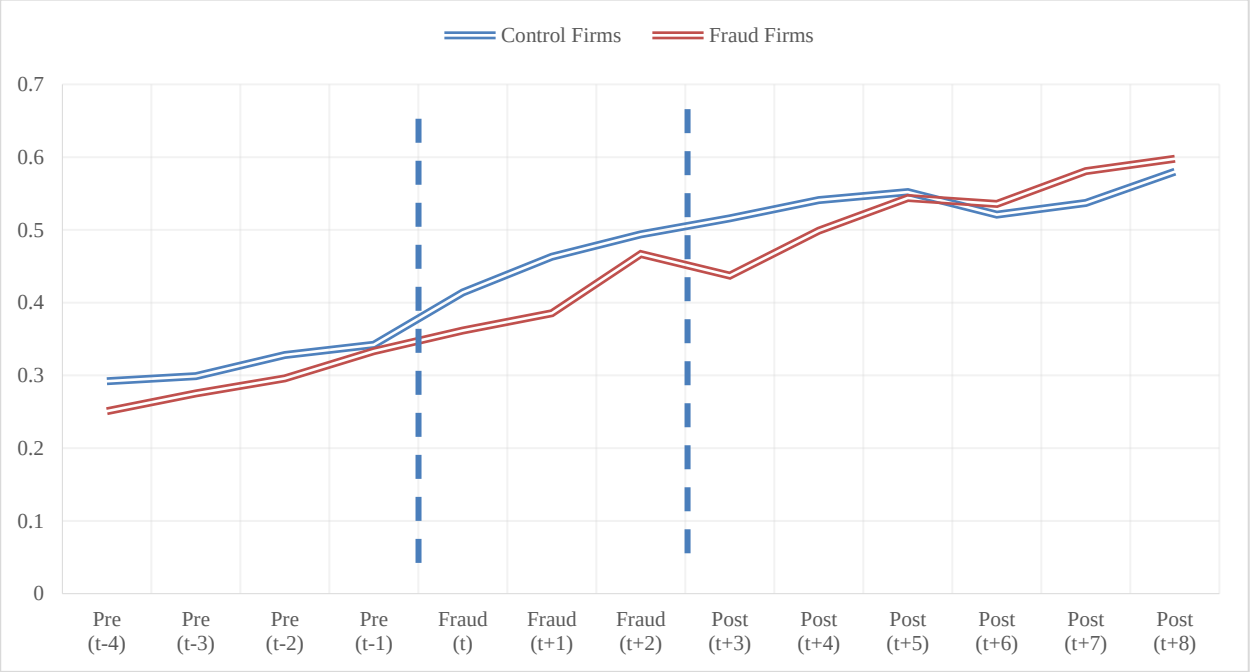
Internet Appendix Figure 3: Manuscript Table 6 Panel C—New Employees

This figure shows data from Table 6 Panel C in the manuscript. Table 6 Panel C shows new employee wages at fraud and matched control firms.



Internet Appendix Figure 4: Firm-wide Tenure in Event Time

This figure shows estimates for firm-wide tenure at fraud firms in the by-event-time years. We sample full-time employees working at a fraud firm or a control firm. The full time workers are defined as a worker at age between 22 and 50 and earning more than $10,000 annually. The long-tenure worker is defined as an employee who has worked for a company more than 2 years. The figure depicts the proportion of long-tenure workers around fraud periods for a fraud firm and a control firm.



Internet Appendix Figure 5: Capital Expenditures in Event Time

This figure shows estimates for capital expenditures scaled by assets at fraud firms and industry averages in the by-event-time years. This sample includes all AAER firms matched to Compustat, not only those from our employee-matched sample from our main analyses in the manuscript. Vertical black bar indicates the approximate timing of the AAER date (i.e., the timing that the SEC publishes its enforcement release) using the median difference between "violation period" and AAER date from Karpoff et al. (2017): 1,008 days after the end of the fraud period.

